# Automation and Inequality Within Firms: Evidence from the Manufacturing Sector in South Africa*

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**Abstract**

This paper investigates the effects of automation on employment dynamics and within-firm earnings inequality in South Africa's formal manufacturing sector. Leveraging matched employer-employee administrative records linked to firm-level import transactions, we find that automation causes a reduction in employment in automating firms relative to those that have not automated yet. However, broadening the control group to include firms that never automate, we find that automating firms are typically on a strong growth trajectory prior to adoption and continue to expand thereafter. This suggests that automation could be associated with a relative deceleration in employment growth rather than with job losses. Overall, average earnings show a modest decline post-automation, while within-firm inequality rises temporarily, although estimates are not always statistically significant. These findings suggest that automation may contribute to rising earnings dispersion within firms, especially in contexts of more substantial technological upgrading.

**Keywords:** automation; technological change; structural change; within-firm inequality; firm-level employment; wage dispersion

**JEL classification codes:** D25; D31; L25; J23; O33

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# 1 Introduction

This paper investigates the role of automation in shaping employment and wage inequality within firms in South Africa's formal manufacturing sector. South Africa is widely recognized for exhibiting some of the highest levels of income inequality globally. This extreme and persistent inequality is rooted in a long history of structural exclusion shaped by colonialism and apartheid and remains one of the most pressing and complex policy challenges in the country.

A substantial empirical literature shows that labour market inequality is the primary driver of income inequality in South Africa, resulting from both a large share of the working-age population being unemployed and a highly unequal distribution among those employed (Bhorat et al., 2020, 2021; Díaz Pabón et al., 2021; Finn et al., 2016; Köhler & Bhorat, 2023; Leibbrandt & Díaz Pabón, 2021; Leibbrandt et al., 2012, 2021; Wittenberg, 2017). Of these, wage disparities among employed workers play a marginally larger role, partly because wages constitute the primary source of household income (Bhorat et al., 2021; Leibbrandt et al., 2012). Given that employees account for about 85% of the workforce, wage inequality unsurprisingly emerges as a major contributor to overall income inequality in South Africa (Díaz Pabón et al., 2021).[1]

Wage inequality, in turn, reflects differences in both worker and firm characteristics. While worker characteristics have been shown to represent the primary driver in developing countries (Messina & Silva, 2019; Rodríguez-Castelán et al., 2022), recent research highlights the importance of firms in driving wage inequality. For example, Alvarez et al. (2018) show that between-firm wage differences accounted for 40% of the decline in earnings inequality between 1996 and 2012 in Brazil. In South Africa, Foster (2023) estimates that worker and firm characteristics explain 35% and 18% of wage inequality, respectively, while Bassier (2023) finds an even larger firm component of 28%. While this emphasises the importance of between-firm differences, differences between workers within the same firm are increasingly recognized as not only a major source of overall wage inequality, but the *dominant* force. Recent decomposition analyses consistently show that the majority of wage inequality arises from within-firm inequality in the country, across different measures of inequality (de la Vega, 2025; Foster, 2023).

Several mechanisms contribute to within-firm wage inequality. For instance, firms often employ workers across a wide range of occupations ranging from low-wage administrative or manual roles to high-wage professional and managerial positions. Additionally, differences in pay structures, such as managerial compensation policies, can create further variation in earnings within the same organization.

Technological change, particularly automation, represents another important driver of within-firm inequality. Technological change, while a fundamental driver of productivity growth, is inherently disruptive. New technologies reallocate resources, displace production processes, and transform labour markets, creating both winners and losers. For firms, automation can lower costs, improve output quality, and support expansion. For workers, effects can be uneven: those whose tasks are complemented by technology may benefit with higher productivity and wages, while those whose tasks are substituted may face displacement or wage stagnation. Existing evidence suggests that automation affects within-firm inequality. For instance, Acemoglu and Restrepo (2020) and Bessen et al. (2023) argue that automation reshapes task allocation and bargaining power in ways that affect the internal distribution of rents. Similarly, Domini et al. (2022) show that robots tend to substitute for low-wage workers but complement high-wage workers.

[1] Over time, wage inequality has changed only marginally, remaining extremely high throughout the post-apartheid period (Jacobs et al., 2024; Köhler, 2024; Shifa et al., 2023). Recent evidence shows that wage inequality drives overall income inequality not only across the broader distribution, but even among top earners (Bassier, 2019; Jacobs et al., 2024). These findings underscore the importance of understanding wage-setting mechanisms and their contribution to inequality in the country.

This paper contributes to the emerging empirical literature on automation and inequality by investigating how firms' investments in automation technologies affect employment and within-firm wage inequality. We focus on the manufacturing sector, where processes are highly standardized and repetitive, making it suitable for robotics and other automation technologies that can deliver measurable improvements in efficiency and cost reduction. Our empirical strategy builds on Bessen et al. (2023) and Domini et al. (2022) and identifies episodes of automation using data on firm-level import transactions – specifically, spikes in imports of capital goods strongly associated with automation. Because spikes occur at different points in time for different firms, we are able to compare outcomes of automating firms to those of firms that haven't automated yet. We do so by applying Callaway and Sant'Anna (2021)'s staggered, heterogeneity-robust Difference-in-Differences estimator to account for variation in automation timing. We apply this approach on rich tax administrative matched employer-employee data covering the entire formal manufacturing sector in South Africa. This context is particularly valuable due to the country's extreme and persistent wage inequality together with the manufacturing sector's key role in industrial development.

Our results indicate that automation causes a relative reduction in employment for automating firms when compared to those that have not automated yet. This relative reduction is compatible with either job losses or growth slowdown, and descriptive evidence points suggests the latter. Including firms that do not automate in the control group disallows causal conclusions but shows how automating firms demonstrate strong employment growth prior to adoption when compared to the average manufacturing firm, a growth that plateaus after automation. Average earnings exhibit a mild reduction post-treatment, although estimates are not always significant. On within-firm wage inequality, results are more nuanced and appear to depend on both automation intensity and firm size. Our baseline estimates suggest modest, temporary increases in inequality, but these effects become more pronounced and statistically significant among firms with larger import spikes and when small firms are excluded. Together, these findings imply that the distributional impacts of automation are concentrated in larger firms undertaking substantial technological upgrades.

Our research contributes to two main strands of the existing literature. First, it adds to the growing body of work on the effects of automation and robotization on labour market outcomes. While early studies focused on aggregate labour market impacts (Acemoglu et al., 2020; Graetz & Michaels, 2018; Michaels et al., 2014), more recent work has shifted attention to firm-level dynamics. In this context, our paper is closely related to Acemoglu et al. (2020), Bessen et al. (2023), Koch et al. (2021), Domini et al. (2021), and Kilumelume et al. (2025). Acemoglu et al. (2020) examine firm-level robot adoption in France and find that adopters increase employment, but this expansion comes at the expense of non-adopting competitors, resulting in a negative net employment effect. Domini et al. (2021), also using French data, identify automation through spikes in capital imports and show that such spikes are associated with an increase in firm net employment growth, driven by both higher hiring and lower separation rates. Koch et al. (2021) analyze Spanish manufacturing firms and find that robot adoption leads to sizeable output gains and net job creation, while also reducing labour cost shares – effects which are concentrated among more productive firms, consistent with positive selection into adoption. Bessen et al. (2023), using Dutch microdata and direct measures of automation expenditures, find that automation raises the likelihood of worker separations which leads to cumulative wage income losses, particularly for older and middle-educated workers as well as those working in smaller firms.

In the case of South Africa, Kilumelume et al. (2025) use similar data to investigate the effects of automation on firm performance and labor market outcomes in manufacturing in South Africa. Their results show that automation adoption leads to substantial increases in firm productivity, value added, and em-

ployment, with positive effects observed across all skill levels within automating firms. We complement their findings by demonstrating that these gains can coexist with rising within-firm wage inequality, highlighting an important distributional dimension of automation. While Kilumelume et al. (2025) document post-automation increases in employment levels relative to non-automating firms, our results show that, conditional on selection into automation, employment growth decelerates after adoption, implying slower expansion rather than job destruction.

Second, our paper contributes to the literature on the determinants of within-firm wage inequality, both globally and in South Africa in particular. On the latter, our findings contribute to the large existing literature on the nature and determinants of wage inequality in the country (Bhorat et al., 2021; Díaz Pabón et al., 2021; Finn et al., 2016; Jacobs et al., 2024; Köhler, 2024; Köhler & Bhorat, 2023; Leibbrandt et al., 2012; Wittenberg, 2017). In particular, it complements Bassier (2023), Foster (2023), and de la Vega (2025) who highlight the role of firms in explaining wage dispersion. It also resonates with Bhorat et al. (2023) who document a pattern of de-routinisation during the post-apartheid period, reflecting broader structural shifts in the labour market driven by technological change. Work in other contexts has begun to explore the role of technological change in shaping wage differentials within firms. Domini et al. (2022) show that robot adoption in France disproportionately benefits higher-wage workers, leading to increased wage inequality within adopting firms. Similarly, Acemoglu et al. (2020) and Bessen et al. (2023) argue that automation can reallocate tasks and bargaining power in ways that alter the internal distribution of rents. Our study adds to both of these literatures by providing new evidence from a developing country context, showing that the distributional consequences of automation – particularly on inequality within firms – can be shaped by firm size and automation intensity.

The remainder of the paper is structured as follows. Section 2 outlines the data sources, construction of within-firm inequality and automation measures, and the empirical strategy. Section 3 presents descriptive statistics and examines the patterns of automation-related capital imports. Section 4 reports the main results on the effects of automation on firm-level employment, average earnings, and within-firm wage inequality. Section 5 concludes.

# 2 Data and method

## 2.1 Data

Our analysis uses merged, annual firm- and employee-level tax administrative panel data covering the universe of formal sector firms and workers in South Africa from 2014 to 2021. The data was made available at South Africa's National Treasury Secure Data Facility (NT-SDF), in the context of the Southern Africa – Towards Inclusive Economic Development (SA-TIED) programme. SA-TIED is a collaboration between the National Treasury; South Africa's tax collection authority, the South African Revenue Service (SARS); and the United Nations University World Institute for Development Economics Research (UNU-WIDER).

All registered firms in South Africa, with few exceptions[2], are required to submit yearly corporate income tax returns using the so-called ITR14 forms[3], which collect detailed financial and operational information, including total sales, cost of sales, and other key performance metrics. Firms are also required to file job-level tax certificates for all employees earning above ZAR 2,000 within a tax year. This is done via IRP5 forms when employee tax is deducted, or IT3(a) forms when not. These forms collect information

[2]These include sole proprietors and partnerships, trusts, and micro businesses who are subject to different tax regimes and filing requirements.

[3]These substituted the older IT14 forms in May 2013.

such as employee earnings, taxes, deductions, and other relevant payments made by the firm (Ebrahim et al., 2021; Pieterse et al., 2018).

The data from each of these tax return forms are structured into distinct, linkable datasets within the NT-SDF. Using employee and firm identifiers, we merge firm-level data from what is referred to as the CIT-IRP5 panel (National Treasury and UNU-WIDER, 2023a) and job-level data from the IRP5 dataset (National Treasury and UNU-WIDER, 2023d). Additionally, we use the employee identifiers created in the individual panel (National Treasury and UNU-WIDER, 2023c) to convert the job-level data to the level of individual workers (Ebrahim & Axelson, 2019). We combine these to create a matched employer-employee dataset which includes data on employee earnings (salaries and benefits).

We further use customs data on firms' import transactions, including product codes, transaction values, and other characteristics (Pieterse et al., 2018). The customs dataset (National Treasury and UNU-WIDER, 2023b) contains firm identifiers that enable us to link import information to our main employer-employee dataset. We restrict the data to include only firms whose primary economic activity is classified within the manufacturing sector, as defined by divisions 10 to 33 of the Standard Industrial Classification (Statistics South Africa, 2012). Following this restriction, together with several cleaning and merging procedures outlined in more detail in Appendix A, our full sample includes 143,779 firm-year observations comprising 32,457 unique firms.

## 2.2 Measures of within-firm inequality

To measure within-firm inequality, we rely on detailed data on the earnings of all employees within each firm. Our primary measure of inequality is the generalised entropy with parameter $\theta = 1$, GE(1), also commonly referred to as the Theil-T index. Generalised entropy captures how a set of values is dispersed from its mean. For $n$ individuals with earnings $w_i$, $i = 1, \ldots, n$, and mean earnings $\mu$, the generalised entropy index with parameter $\theta = 1$ is given by:

$$GE(1) = \frac{1}{n} \sum_{i=1}^{n} \left[ \frac{w_i}{\mu} \cdot \ln \left( \frac{w_i}{\mu} \right) \right].$$

Generalized entropy measures are advantageous in that they can be additively decomposed into within- and between-group components. Prior research demonstrates that the within-sectors-within-firms component accounts for the largest share of overall GE(1) inequality in South Africa's formal sector (de la Vega, 2025), motivating our use of this measure as this study's primary indicator of within-firm inequality. Our use of GE(1) also facilitates a comparison of our main results to what is happening at different points of the within-firm distribution. Changing the parameter from 1 to 0 or 2, for instance, makes the measure more sensitive to the bottom or the top of the distribution, respectively. We perform robustness checks changing the inequality measure to GE(0) and to GE(2). The generalised entropy index with parameter $\theta$ is:

$$GE(\theta) = \frac{1}{n} \frac{1}{\theta(\theta - 1)} \sum_{i=1}^{n} \left[ \left( \frac{w_i}{\mu} \right)^{\theta} - 1 \right].$$

## 2.3 Methodology

To identify the effects of automation on firm growth and inequality, we employ a heterogeneity-robust, staggered Difference-in-Differences (DiD) design, investigating the outcomes of firms that adopt automation technologies while accounting for variation in automation timing. We follow the literature (Bessen

et al., 2023; Domini et al., 2021, 2022) in proxying firm-level automation by the imports of capital goods typically used for the automation of manufacturing processes as identified in Acemoglu and Restrepo (2022) and listed in Table 1.

Hence, our empirical strategy relies on two additional key assumptions alongside the standard DiD identifying assumptions of no-anticipation and parallel trends. First, that South African manufacturing firms automate by importing such capital goods. Naturally, any acquisition of such products e.g. via local third parties or brokers will not be visible in our approach and thus constitutes a limitation of our study. Second, that automation investments occur in discrete spikes rather than gradually, which is consistent with broader firm-level investments patterns observed in other contexts (Bessen et al., 2023; Domini et al., 2021, 2022).

Table 1: Capital goods related to automation in manufacturing.

| Product | HS-2012 codes (6 digits) |
|---|---|
| Industrial robots | 847950 |
| Dedicated machinery | 847989 |
| Automatic machine tools | 845600–846699, 846820–846899, 851511–851519 |
| Automatic welding machines | 851521, 851531, 851580, 851590 |
| Weaving and knitting machines | 844600–844699, 844700–844799 |
| Other textile dedicated machinery | 844400–844590 |
| Automatic conveyors | 842831–842839 |
| Automatic regulating instruments | 903200–903299 |
| 3-D printers | 847780 |
| Automatic data processing machinery | 847141–847150, 847321, 847330 |
| Electronic calculating machines | 847010–847029 |

Source: Acemoglu and Restrepo (2022).

Our baseline approach to identify automation spikes follows Domini et al. (2021, 2022) and defines a spike year for each individual firm as the year in which it has the highest value of automation-related imports. All other years are treated as non-automation years, regardless of whether import values for the capital goods of Table 1 are positive. While this method is straightforward, it may of course understate the extent of automation if firms adopt these technologies either gradually or in multiple instances. As such, we perform robustness checks using two alternative approaches to identify automation spikes. The first, also used by Domini et al. (2021, 2022), addresses the limitations of our baseline approach by keeping only firms with a single automation *peak*, where peak years are defined as those in which a firm's imports of automation-related capital goods exceed 20% of the value in its highest import year. Firms with multiple peaks are excluded from the analysis. For the remaining firms, the year with the highest import value is retained as its automation spike. The second alternative approach, adapted from Bessen et al. (2023), applies a more stringent criterion to identify automation spikes. Rather than relying solely on import values, it considers the ratio of automation-related capital imports to total operational costs. The spike year is then identified as the year within which this ratio exceeds three times the firm-specific average. As with the previous definition, we exclude firms with multiple spikes from the sample. Hence, one drawback of these alternative approaches is the reduction in sample size due to the exclusion of firms with repeated automation episodes. The first alternative definition cuts the number of treated firms by nearly half, and the second definition by nearly a sixth.[4]

Once treated firms are identified, we apply the staggered, semi-parametric, heterogeneity-robust DiD estimator by Callaway and Sant'Anna (2021) to isolate the average and dynamic causal effects of au-

[4]Table 4 in the next section gives a more detailed account of the number of treated firms by year of treatment for each approach.

tomation on firms who automated. A new theoretical and empirical literature on the econometrics of DiD has developed over the last few years, highlighting how many applications of this method do not meet all requirements outlined in the canonical two-group two-period setup. This is primarily an issue in the presence of heterogenous or 'staggered' treatment timing, when the event being studied does not occur at only one point in time (Baker et al., 2025; Callaway & Sant'Anna, 2021; Goodman-Bacon, 2021; Roth et al., 2023; Sun & Abraham, 2021). A standard DiD design applied to such setting can yield biased estimates due to the inclusion of both 'clean' comparisons (between treated and not-yet- or never-treated units) as well as 'forbidden' comparisons (between units who are both already-treated but in varying periods). This issue is particularly relevant in our context as firms adopt automation in different years, and the effects of automation may vary across time and firm types. For example, early adopters may differ systematically from late adopters in terms of size, sector, or pre-trends in employment and earnings.

The intuition behind the Callaway and Sant'Anna (2021) estimator is that only firms who have either never and/or not-yet automated in a given period are included in the control group. The key concept behind this approach is the group-time average treatment effect, $ATT(g,t)$, where group $g$ indexes the time period that firms first automate and $t$ a given time period. This can be formally expressed as follows:

$$ATT(g,t) = [EY(g)_t - EY(C)_t] - [EY(g)_{g-1} - EY(C)_{g-1}] \tag{1}$$

where $EY(g)_t$ is the mean value of a given outcome for group $g$ at time $t$, and $EY(C)_t$ the mean value of the outcome for the control group $C$ in the same period. In our analysis, we focus on three sets of outcomes at the firm-level: total employment (log scale), average log earnings, and the within-firm inequality indices defined earlier. The first term calculates the between-group difference in outcomes at time $t$, while the second term does the same but at time $g-1$ which corresponds to the period immediately preceding group $g$'s first automation period. Hence, the estimator accounts for variation in treatment timing by separately calculating the ATT for all groups across all periods. Because estimation for each group $g$ requires at least one period of pre-treatment data, firms who are 'always-treated' are automatically omitted from the sample. Lastly, this estimator assumes that automation is an 'absorbing state'; that is, it is irreversible such that automation exposure is 'weakly increasing'. We believe this to be an appropriate assumption given that it implies firms do not 'forget' about their automation history.[5]

The estimator's identifying assumptions include *parallel trends* – which implies that in the absence of automation, the average trend in a given outcome would have been the same for automating and non-automating firms in all post-treatment periods – and *no anticipation* – which implies that effects are zero in all pre-treatment periods due to eventual automating firms not responding on the potential knowledge of automating in the future. We test for the plausibility of these assumptions through the use of aggregations. Given the presence of staggered treatment timing and hence multiple $g$ cohorts, Equation (1) implies that this estimator produces numerous $ATT(g,t)$'s, as opposed to the single ATT in canonical designs, which may be cumbersome to report. We give focus specifically to the event study aggregation, which allows us to study effect dynamics or, equivalently here, variation in the number of years firms had automated for. It also allows us to obtain pre-treatment effect estimates to gauge the plausibility of our identifying assumptions.[6] For each event time $e = t - g$, these are calculated as the

[5] An alternative estimator by de Chaisemartin and D'Haultfœuille (2023) does allow for treatment to turn on and off, however only subject to a strict 'no carryover' assumption which imposes that potential outcomes only depend on current treatment status and not on full treatment histories (Roth et al., 2023). Given the possibility of dynamic and cumulative effects of automation, we believe this assumption is inappropriate in this context and thus proceed with Callaway and Sant'Anna (2021)'s approach.

[6] In doing so, we estimate "long differences" for both the pre- and post-treatment coefficients to ensure they are numerically equivalent to the dynamic two-way fixed effects (TWFE) specifications in a non-staggered setup (Roth et al., 2023). In other words, we always use the period immediately before treatment as the baseline. As shown by Roth et al.

weighted average of each $ATT(g, g+e)$ with weights that are proportional to the treatment group size. The derivation of the event study aggregation is formally described in Callaway and Sant'Anna (2021).

There is reason to believe firms could self-select into treatment, due to the nature of treatment and confirmed in descriptive evidence shown in Section 3. To account for this, estimates for a given event time $e$ are derived by comparing firms that experience an automation spike with firms that have not yet automated, but will do so in the future. Restricting the control group to not-yet-treated firms enhances causal identification by improving the comparability between treated and control firms, thereby reducing bias from selection and any unobserved confounders. Results including never-treated firms are used as complementary descriptives.

We also perform a series of robustness checks. Inequality measures may be distorted in firms with very few employees, and more sensitive to individual outliers. Small firms may also differ systematically in their capacity to invest in automation. For those reasons, our first robustness check excludes firms with 10 or fewer employees from the analysis. As mentioned above, our second robustness check is to adopt alternative, more demanding approaches to identify spikes, and our third one is to check results for inequality as measured by GE(0) and GE(2) instead of GE(1). The fourth robustness check follows Kilumelume et al. (2025) in performing a placebo test by assigning treatment randomly to firms. We select the number of treated firms in each year based on the observed number of treated firms as shown on Table 4. Because our core results are sensitive to firm size, we disaggregate the numbers of Table 4 in three size ranges (10 workers or less; 11–100 workers; and more than 100 workers) and preserve the number of treated firms in each year and size category. We also explore the role of heterogeneity in treatment intensity by separately analysing firms with value of imports at the spike above and below the median, and for large firms and SMEs one at a time. We cluster standard errors at the panel (firm) level and report these throughout the results. As a final robustness test, we re-estimate them using a multiplicative wild bootstrap procedure with 1,000 replications, and report cases in which statistically significant coefficients discussed in the text become non-significant under the bootstrap.

# 3 Descriptive statistics

The main sample is restricted to the universe of formal manufacturing firms from 2014 to 2021. Table 2 presents the total number of firms by year, distinguishing between those that appear only in the firm-level dataset and those that are also present in the customs (imports) data. The number of firms remains relatively stable between 2014 and 2019, followed by a decline in 2020 and particularly in 2021, which may reflect the net impact of the COVID-19 pandemic and associated restrictions. Using labour force survey data, Kohler et al. (2022) show that South Africa's secondary sector – particularly manufacturing and construction – was disproportionately impacted by the pandemic, accounting for nearly one-third of all net job losses alone.

Non-importing firms serve as the never-treated subset of our control group firms. Importing firms may fall into one of three categories. First, because import data is available from 2009 to 2022, it is possible that firms had their automation spike prior to 2014, the initial period of firm-level data used in this paper. Importing firms with an automation spike in 2014 or earlier are the always-treated subset of our treated firms. These firms are included in the descriptives under this section but are dropped from the DiD analyses later on, as mentioned in the preceding section. The second category is that of firms with

(2023), the default Callaway and Sant'Anna (2021) event study estimates are constructed asymmetrically for the pre- and post-treatment periods, such that the former comprise comparisons of consecutive periods (referred to as "short differences") while the latter are "long differences". This asymmetry can result in a spurious jump in the plot even if there is no treatment effect.

Table 2: Total observations per tax year of manufacturing firms by importing status

| Tax year | Importing | Non-importing | Total |
|---|---|---|---|
| 2014 | 2,075 | 16,372 | 18,447 |
| 2015 | 2,041 | 16,425 | 18,466 |
| 2016 | 2,042 | 16,519 | 18,561 |
| 2017 | 2,158 | 16,432 | 18,590 |
| 2018 | 2,100 | 16,254 | 18,354 |
| 2019 | 2,098 | 16,576 | 18,674 |
| 2020 | 1,829 | 16,378 | 18,207 |
| 2021 | 1,405 | 13,075 | 14,480 |
| Total | 15,748 | 128,031 | 143,779 |

Source: authors' own calculations.

a spike between 2015 and 2022, which form the treated groups in our analyses. Since firm-level data only goes up to 2021, firms with a spike in 2022 are only used as control group for their outcomes before treatment due to our staggered treatment design. Finally, it is possible that importing firms do not have a single identifiable automation spike. While this cannot occur using our main definition of spikes, it can occur for the alternative definitions we use in our robustness checks. Such firms are referred to as having a 'smoothed' automation behaviour in the descriptives of this section and are also removed from the event studies.

We turn to the characterisation of imports of capital goods related to the automation of manufacturing jobs. Key capital goods account for approximately 1.6% to 2.1% of total imports realised in the South African economy between 2009 and 2022. Approximately 9,000 to 10,000 firms (20–23% of all firms) import these goods during a given year. As such, the share of firms which import these goods is significantly higher than the value of these imports for every year. Over time, while the share of firms who import these goods has remained relatively constant, these imports as a share of total imports have declined. However, because this data uses current prices, it is possible that these trends are being driven by changes in relative prices of the key capital goods in relation to the overall economy. Table C1 in the Appendix reports the number of importing firms and total import value for these key capital goods and for all imported products, across the South African economy (not just in the manufacturing sector).

Table 3 presents trends in the total value of imports of automation-related capital goods between 2014 and 2021 for each category listed in Table 1. Most imports that characterise the automation spikes in our analyses are concentrated in four products, which account for 83% of the total value of imports of automation-related capital goods. In descending order, these include automatic data processing machinery (42%), automatic machine tools (20%), dedicated machinery (13%), and automatic regulating instruments (8%). In the Appendix, Table C4 shows the same data by HS4 code, whose descriptions are given in Table C5.

We now examine import patterns among automating firms, i.e., manufacturing firms that imported these automation-related capital goods. Figure 1 shows the distribution of firms by the number of years in which they imported key capital goods between 2014 and 2021. Most firms import these goods in only one year during the period. Despite this, multiple firms continue to import in multiple years, motivating our use of alternative approaches to identify automation spikes as robustness checks, as these methods are better suited to cases where firms import in multiple periods.

Table 4 presents, for each of the three definitions of automation spikes, the number of unique firms by treatment year, together with the accompanying number of never-treated and always-treated firms, and

Table 3: Yearly imports ($10^9$ ZAR) per category of key capital goods

| Category | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | Total | Share |
|---|---|---|---|---|---|---|---|---|---|---|
| Automatic data processing machinery | 6.89 | 8.21 | 8.35 | 7.68 | 9.65 | 12.02 | 8.97 | 8.62 | 70.38 | 42.3% |
| Automatic machine tools | 3.93 | 4.48 | 4.37 | 4.18 | 4.08 | 3.76 | 3.31 | 4.97 | 33.07 | 19.9% |
| Dedicated machinery | 3.07 | 3.23 | 2.36 | 2.30 | 2.97 | 2.37 | 2.84 | 2.35 | 21.49 | 12.9% |
| Automatic regulating instruments | 1.55 | 1.69 | 1.72 | 1.50 | 1.67 | 1.58 | 1.54 | 1.73 | 12.98 | 7.8% |
| Automatic conveyors | 0.62 | 0.55 | 0.72 | 1.03 | 1.03 | 1.62 | 1.74 | 1.01 | 8.32 | 5.0% |
| Automatic welding machines | 0.63 | 0.66 | 0.95 | 0.91 | 0.63 | 1.33 | 0.79 | 1.88 | 7.78 | 4.7% |
| 3-D printers | 0.56 | 0.57 | 0.50 | 0.68 | 0.50 | 0.52 | 0.55 | 0.37 | 4.25 | 2.6% |
| Industrial robots | 0.13 | 0.21 | 0.66 | 0.40 | 0.52 | 0.59 | 0.32 | 0.67 | 3.50 | 2.1% |
| Weaving and knitting machines | 0.21 | 0.37 | 0.19 | 0.27 | 0.31 | 0.32 | 0.16 | 0.22 | 2.05 | 1.2% |
| Electronic calculating machines | 0.17 | 0.17 | 0.16 | 0.17 | 0.20 | 0.20 | 0.16 | 0.16 | 1.40 | 0.8% |
| Other textile dedicated machinery | 0.11 | 0.12 | 0.31 | 0.12 | 0.25 | 0.12 | 0.13 | 0.10 | 1.26 | 0.8% |

Source: authors' own calculations.

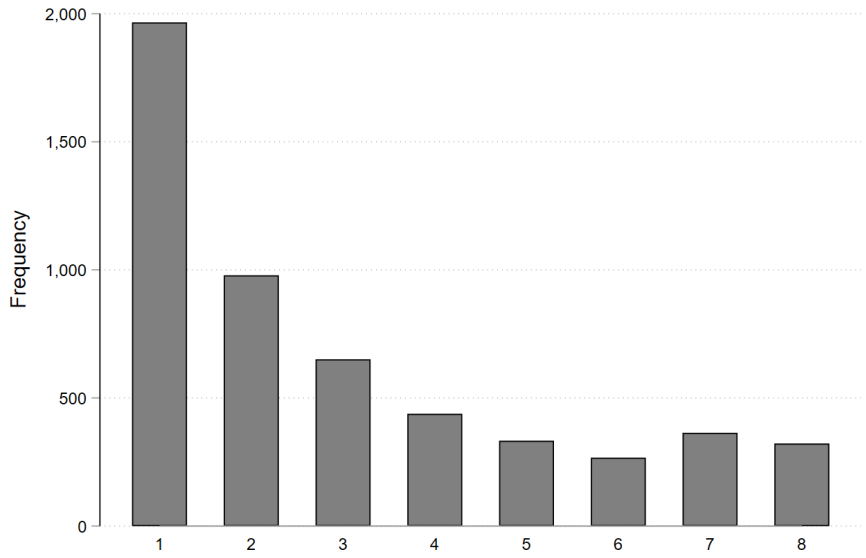



Figure 1: Count of manufacturing firms by number of years with imports of key capital goods in 2014–2021
Source: authors' own calculations.

those with 'smoothed' automation[7]. We also show the total number of firm-year observations for each of these categories. We observe a relatively stable distribution of treated firms in terms of the timing of their imports, except for a drop coinciding with the COVID-19 pandemic in 2020. This suggests that our effect estimates are unlikely to be driven by a single cohort of firms dominating the overall estimated effects. As expected, we also observe a smaller number of firms under the stricter definitions of automation spikes.

In Table 5 we present summary statistics for selected variables for our analytical sample. Since these variables often have skewed distributions and some degree of persistence, we expect more variation across firms than within firms over time. Consistent with this expectation, the between-firm standard deviations exceed the within-firm standard deviations for every variable. The panel is unbalanced and firms are present on average for 4.4 out of the 8 years during the study period. A closer look at firm participation over time shows that the most common pattern is for firms to be present during all eight years, which is the case for only 7,227 (21%) firms. This is followed by 2,178 (6.3%) firms that are present only in 2014,

[7]As discussed, always-treated firms include those who experience an automation spike before 2015, while 'smoothed' firms are those with multiple automation spikes. Both are excluded from the analyses. When using the definition of spike borrowed from Bessen et al. (2023), peaks and spikes can only be identified between 2014 and 2021, since that definition uses firm-level data on costs.

Table 4: Unique manufacturing firms and total observations by year of treatment for each approach.

| Year of treatment | Firms | | | Observations | | |
|---|---|---|---|---|---|---|
| | Main[a] | RC 1[b] | RC 2[c] | Main | RC 1 | RC 2 |
| Never-treated | 25,727 | 25,702 | > 31,700[g] | 105,786 | 105,753 | 139,045 |
| 2009[d] | 435 | 232 | -[e] | 2,277 | 1,195 | - |
| 2010[d] | 369 | 190 | - | 1,977 | 994 | - |
| 2011[d] | 456 | 227 | - | 2,443 | 1,205 | - |
| 2012[d] | 503 | 236 | - | 2,660 | 1,239 | - |
| 2013[d] | 514 | 241 | - | 2,773 | 1,193 | - |
| 2014[d] | 469 | 212 | 82 | 2,515 | 1,103 | 604 |
| 2015 | 462 | 192 | 98 | 2,720 | 1,097 | 691 |
| 2016 | 469 | 196 | 96 | 2,816 | 1,077 | 682 |
| 2017 | 493 | 237 | 74 | 3,005 | 1,381 | 530 |
| 2018 | 510 | 241 | 72 | 3,046 | 1,330 | 506 |
| 2019 | 550 | 258 | 76 | 3,253 | 1,472 | 554 |
| 2020 | 405 | 189 | 82 | 2,380 | 1,056 | 586 |
| 2021 | 504 | 210 | 73 | 2,878 | 1,158 | 557 |
| 2022 | 591 | 270 | - | 3,250 | 1,392 | - |
| Smoothed[f] | - | 3,624 | *[g] | - | 21,134 | 24 |

Notes: (a) Refers to the main approach (Domini et al., 2022) employed for identifying spikes as detailed in section 2.3. (b) Refers to the alternative approach in the first robustness check (Domini et al., 2022). (c) Refers to the alternative approach in the second robustness check (Bessen et al., 2023). (d) All firms with treatment in 2014 or before are considered always-treated and thus excluded from the analyses in Section 4. (e) Because the Bessen et al. (2023) approach uses firm-level data, spikes are only identified between 2014 and 2021. (f) Refers to firms with more than one peak in the definitions of the robustness checks, which are excluded from the analyses. This cannot occur under the main approach. (g) Number of firms with smoothed behaviour below 10 and thus omitted to comply with rules for data extraction. Precise number of never-treated firms removed accordingly to avoid reconstructing the table by column sums.
Source: authors' own calculations.

and 2,072 (6.0%) firms that are present in all years apart from 2021. Over 80% of firms do not exhibit any gaps in their participation — that is, they are present in all years until their last appearance, or in all years following their first one. Table C3 in the Appendix provides a detailed account of the participation patterns in the unbalanced panel.

Table 5: Summary statistics of selected variables for manufacturing firms

| Variable | Mean | SD | SD (b) | SD (w) | Total obs. | Panels | $\bar{T}$ |
|---|---|---|---|---|---|---|---|
| Generalised entropy, $\theta = 1$ | 0.354 | 0.241 | 0.217 | 0.115 | 143,779 | 32,457 | 4.4 |
| Log of number of workers | 2.839 | 1.346 | 1.303 | 0.325 | 143,779 | 32,457 | 4.4 |
| Average log earnings paid by the firm | 10.982 | 0.737 | 0.720 | 0.312 | 143,779 | 32,457 | 4.4 |
| Generalised entropy, $\theta = 0$ | 0.380 | 0.252 | 0.224 | 0.142 | 143,779 | 32,457 | 4.4 |
| Generalised entropy, $\theta = 2$ | 0.547 | 0.666 | 0.559 | 0.322 | 143,779 | 32,457 | 4.4 |

Source: authors' own calculations.

Table 6 provides summary statistics separately for automating firms and those that never did so during the study period, using our main treatment definition. On average, automating firms employ more workers and offer higher earnings. Importantly, they also display substantially higher within-firm earnings inequality compared to non-automating firms. These patterns point to potential self-selection into treatment, which we explore further in the subsequent analysis.

Finally, Table 7 shows time trends for the averages of these firm-level outcomes. As expected, all variables have high permanence through time. In any case, inequality measures drop slightly and continuously since the start of the series, only to revert in 2021 to values close to the initial ones; while the inverse is true for average log earnings. Furthermore, firms show a sustained trend of reduction in size when measured

Table 6: Summary statistics for never-treated and automating firms for the main definition of spikes.

| Variable | Never-treated | | Automating | | Diff. |
|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | |
| Generalised entropy, firm-level ($\theta = 1$) | 0.33 | 0.24 | 0.42 | 0.24 | 0.085*** |
| Log of number of workers | 2.57 | 1.18 | 3.59 | 1.48 | 1.021*** |
| Average log earnings paid by the firm | 10.89 | 0.72 | 11.23 | 0.72 | 0.340*** |
| Generalised entropy, firm-level ($\theta = 0$) | 0.36 | 0.25 | 0.44 | 0.24 | 0.078*** |
| Generalised entropy, firm-level ($\theta = 2$) | 0.49 | 0.60 | 0.70 | 0.80 | 0.211*** |

Source: authors' own calculations.

as the number of workers, with a further drop in 2021. These results somewhat mirror the trend for the number of firms in the formal economy as shown in Table 2 and once more may be expressions of the impact of the COVID-19 pandemic in South Africa.

Table 7: Yearly cross-firm averages of selected variables for manufacturing firms.

| Tax year | GE(0) | GE(1) | GE(2) | Log workers | Avg. log earnings |
|---|---|---|---|---|---|
| 2014 | 0.392 | 0.367 | 0.564 | 2.862 | 10.980 |
| 2015 | 0.392 | 0.367 | 0.573 | 2.872 | 10.994 |
| 2016 | 0.385 | 0.360 | 0.560 | 2.858 | 10.993 |
| 2017 | 0.378 | 0.353 | 0.546 | 2.842 | 11.003 |
| 2018 | 0.374 | 0.350 | 0.543 | 2.852 | 11.013 |
| 2019 | 0.368 | 0.342 | 0.526 | 2.809 | 11.016 |
| 2020 | 0.363 | 0.336 | 0.512 | 2.812 | 11.022 |
| 2021 | 0.388 | 0.359 | 0.550 | 2.795 | 10.795 |

Source: authors' own calculations.

# 4 Main findings

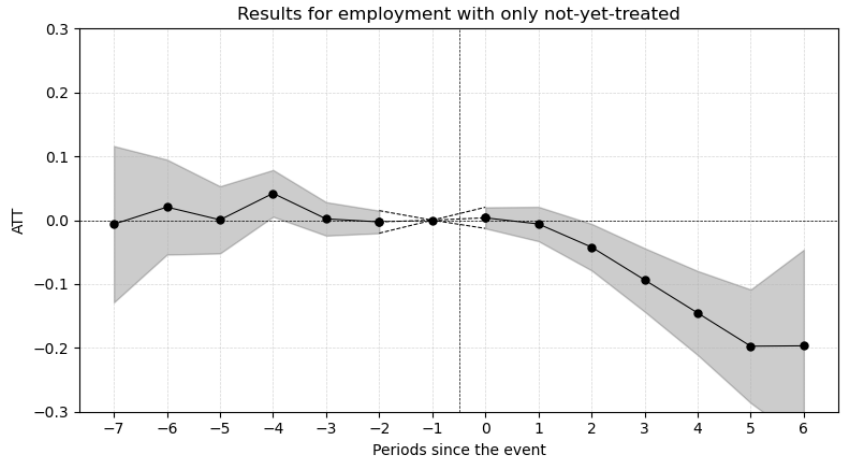
## 4.1 Employment

Figure 2 presents our event study estimates of the average effects of automation, using the logarithm of total employment as the outcome and automation spikes as treatment. The spike year is defined as the year with the highest value of automation-related imports for a given firm, as discussed in Section 2.3. These effects should be interpreted as average treatment effect on the treated (ATT), that is, the average effects of automation on automating firms themselves. To account for self-selection into treatment, the main results exclude never-treated firms from the control group, which is only composed of not-yet-treated firms. Results including never-treated firms are used as descriptives that complement the main results.

We do not find evidence of significant pre-trends in Figure 2, with the exception of the estimate for $t-4$, that is no longer significant under bootstrapped standard errors. Treated and not-yet-treated firms follow similar trends in average employment during the years leading up to automation, supporting the credibility of our identification strategy. Following automation, however, a decline in relative employment among automating firms materialises two years following automation and grows in time. Under bootstrapped standard errors, the decline becomes significant at period $t+3$.

These findings echo those of Bessen et al. (2023), who also use not-yet-treated firms as a comparison group and find similar results for the Dutch manufacturing sector. They suggest that automation could lead to a significant decline in employment among automating firms. However, they are also compatible with a slowdown in growth rather than with job losses. Broadening the control group to also include non-importing manufacturing firms, as shown in Figure B1 in the Appendix, helps understanding what

Figure 2: Automation and Firms' Total Employment - Early vs Late Adopters




Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

happens to firms who do automate. Coefficients in the pre-automation period are consistently negative and significant, which clearly violates the parallel trends assumption and prevents causal identification but still provides useful descriptive insight. It suggests that firms that automate were, on average, smaller during the period leading up to automation, with the gap in number of employees becoming smaller and approaching zero leading up to the automation event. That is, automating firms have experienced faster employment growth during the years just before automation than the average manufacturing firm. Following automation, relative employment continues to grow for two years and then plateaus. This suggests that the reduction seen in Figure 2 is not due to job losses, but rather to a reduction in the rate of hiring after automation. These results suggest that automation spikes tend to occur in growing firms and may be part of a broader scaling-up process rather than a downsizing strategy. This pattern of relative growth in the lead-up to automation when all non-importing firms are included in the control group is even more pronounced among SMEs, as shown in Figure B2 in the Appendix, reinforcing the conclusions.

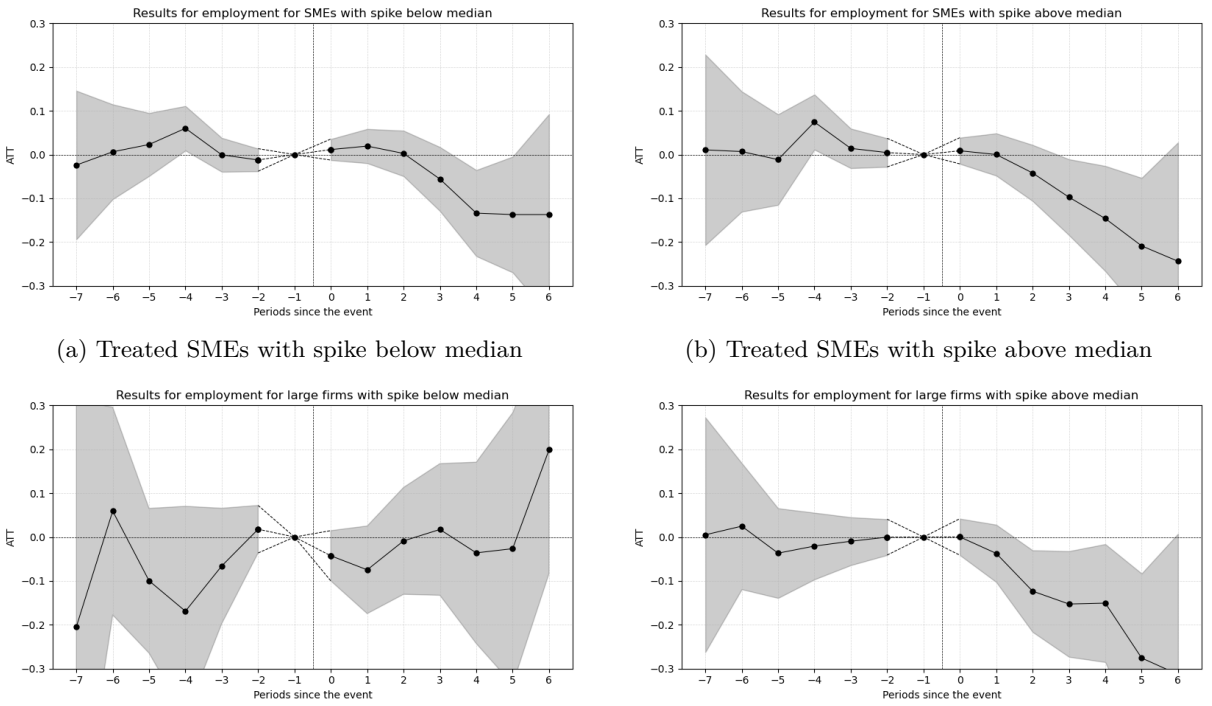
Results including never-treated firms in the control group also add to the evidence on selection into automation, whereby growing firms are more likely to adopt advanced technologies to sustain or accelerate their performance. For instance, Acemoglu et al. (2020) highlight that firm-level technology adoption is often endogenous and reflects a broader strategy of capital deepening and market expansion. Domini et al. (2021) show that automation spikes are positively correlated with preceding and contemporaneous growth in employment. Moreover, Bessen et al. (2023) show that automation tends to be adopted by firms that are already more productive and growing faster than their peers. Our results therefore seem to corroborate those in the existing literature. In addition, our results on employment growth after automation are consistent with Kilumelume et al. (2025), who also find that automation is associated with increased employment in South African manufacturing firms in an analysis that includes both never-treated and not-yet-treated firms.

These results are robust to alternative empirical setups. Figure B3 in the Appendix shows that similar results are found when removing small firms (10 or fewer workers), although post-treatment significance only kicks in at $t + 3$. In a placebo test that assigns treatment randomly, results largely disappear. As shown in Figure B4a and Figure B4b in the Appendix, nearly all estimates before and after treatment

become null, with an exception for negative and significant[8] coefficients at $t+3$ and $t+4$, which could be related to the continued trend of reduction in the average log number of workers per firm as documented in Table 7. While this brings some nuance to our main findings, it doesn't challenge them entirely both because the reduction after the placebo treatment is much less pronounced than the one we see in our core results, and because results with and without never-treated firms are similar, unlike in our analysis.

We also explore differences in treatment intensity by splitting the sample along two dimensions: (a) whether a firm's peak import value falls below or above the median peak import value across all treated firms; and (b) whether the number of workers in the firm's first observation is below 100 or if it is 100 or more. The control group consists of only not-yet-treated firms in all cases. Results for employment, shown in Figure 3, are more pronounced for firms with larger spikes. Large firms with spike below the median do not show an observable effect, although because not a lot of firms fall under this category, there is an important volatility. For small and medium enterprises, both groups show similar dynamics, although firms with large spikes experience post-treatment effects one period earlier and a sustained relative fall in employment, whereas among firms with smaller spikes the decline plateaus by the fourth period after treatment.

Figure 3: Automation and Firms' Total Employment by Automation Intensity – Early vs. Late Adopters




(a) Treated SMEs with spike below median

(b) Treated SMEs with spike above median

(c) Large treated firms with spike below median

(d) Large treated firms with spike above median

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The control group includes only not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.
Source: authors' own calculations.

We run an additional robustness check where alternative, more demanding approaches to identifying spikes are adopted. In the first alternative approach (Domini et al., 2022), all years in which automation-related imports exceed 20% of the firm's peak import year are classified as additional peaks. Firms with a single peak are deemed to have a spike in that year and are considered treated. Firms with multiple

[8]For bootstrapped standard errors, only the estimate at $t+3$ for the control group with never-treated firms remains significant.

peaks are deemed to have a smoothed behaviour and are excluded from the analysis. We focus on results for control groups with only not-yet-treated firms. As previously shown in Table 4, this stricter approach for identifying spikes leads to a smaller number of treated firms, reducing the precision of estimates, leading to less significant results overall. Results for employment are again similar, although pre-trends are less clearly null and post-treatment statistically significant effects emerge only in periods $t + 4$ and $t + 5$, as shown in Figure B5 in the Appendix.

We also employ a second alternative approach to identify spikes which looks not only at the value of imports, but also at the ratio of those over operational costs in each year, following Bessen et al. (2023). It then compares the ratio in each year to the firm-specific average, with values above 3 being considered peaks. Once more, firms with a single peak are considered treated in that year, and firms with multiple peaks are excluded. Results from that second robustness check are less striking. Point estimates are overall in line with the baseline narrative, although with much more muted patterns for employment. However, virtually none of the outcomes explored in our analyses exhibit any significant parameters, neither in the lead-up to treatment nor after it, a notable exception being the behaviour of employment when restricting to small and medium firms and with never-treated firms in the control group. This might be driven by the even smaller number of treated firms that are left after this more strict approach is applied. As shown previously on Table 4, while the number of treated firms falls by half between the main approach and the approach in the first robustness check, it falls six-fold for this second alternative approach. We show the main results in Figure B6 of the Appendix but refrain from exploring them in detail given the lack of significance across estimations.

## 4.2 Earnings and Within-Firm Inequality

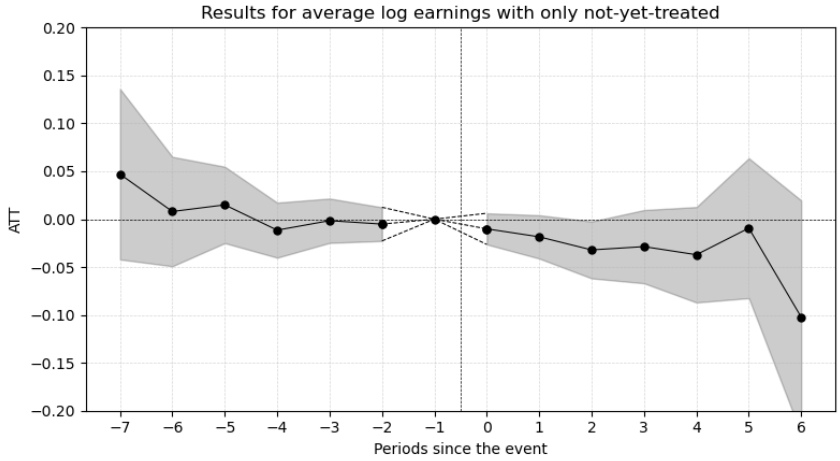
We turn to our estimates of the effects of automation on firms' average earnings and on within-firm earnings inequality. Again, given the evidence for self-selection, we focus on the analyses that keep only not-yet-treated firms as the control group, and mention those including never-treated firms as complementary descriptives. Figure 4 presents the estimates for average earnings as the outcome variable. Pre-treatment estimates are close to zero and insignificant, with no evidence of diverging earnings trajectories prior to automation, reinforcing the credibility of our identification strategy. Following automation, coefficients turn negative, but they are not significant, with the exception of $t+2$, even though that significance goes away when standard errors are bootstrapped.

Figure 5 presents the estimated effects of automation on within-firm inequality, measured as GE(1), with only not-yet-treated firms in the control group. The majority of estimates in the pre-treatment period are close to zero and statistically insignificant, again supporting the parallel trends assumption. Following automation, the point estimates of inequality initially rise until $t + 1$ and later fall, although only the estimate for $t + 1$ itself is significant, and not so under bootstrapped standard errors.

It is worth mentioning that including never-treated firms in the control group does not change results substantially, as shown in the Appendix in Figure B7 for average earnings and Figure B8 for inequality. For these outcomes, we find no evidence of selection into automation, unlike the pattern observed with employment. For average earnings, coefficients again become slightly negative but not significant following automation. For inequality, the significant but temporary rise in inequality is even more pronounced, with point estimates becoming positive and significant during the post-treatment period[9].

[9]Under bootstrapped standard errors, only the estimate for GE(1) at period $t + 1$ remains significant.

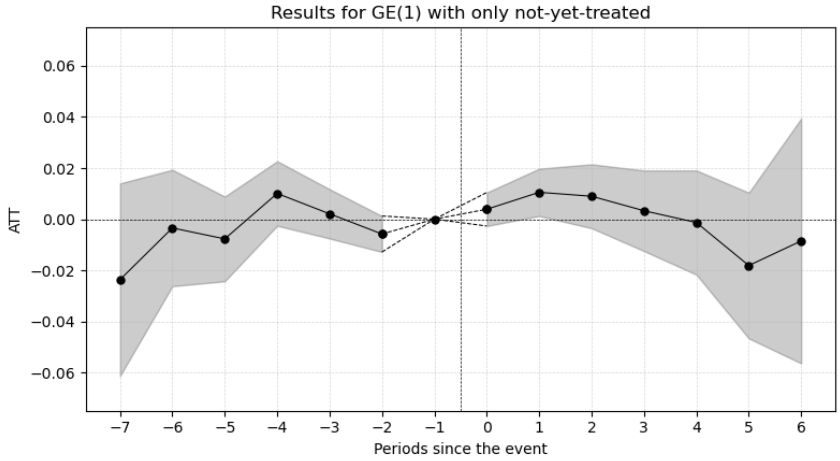
Figure 4: Automation and Firms' Average Earnings - Early vs Late Adopters




Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the firm-level average of the logarithm of earnings. The control group includes only not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

Figure 5: Automation and Within-Firm Inequality - Early vs Late Adopters

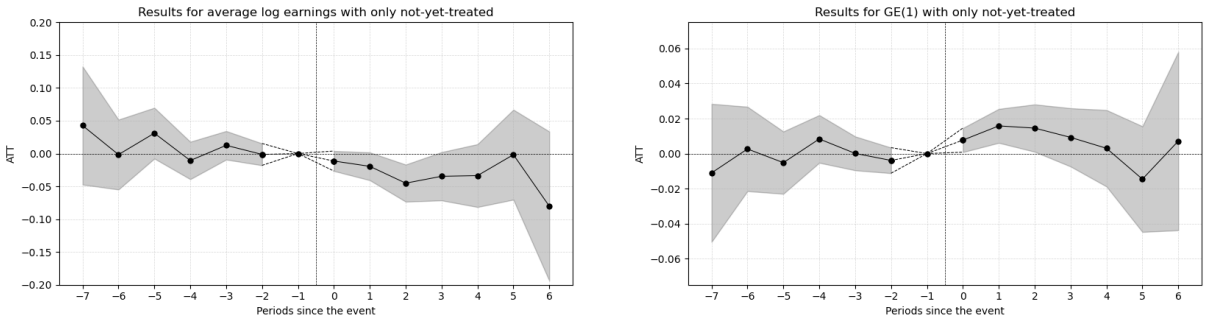



Note: Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is GE(1). The control group includes only not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

When excluding small firms, these results change in two important ways: we now observe a statistically significant[10] decline in firms' average earnings following automation (Figure 6a) and a clearer rise in within-firm inequality compared to the baseline (Figure 6b).

Figure 6: Automation, Earnings, and Within-Firm Inequality - Early vs Late Adopters, Excluding Small Firms



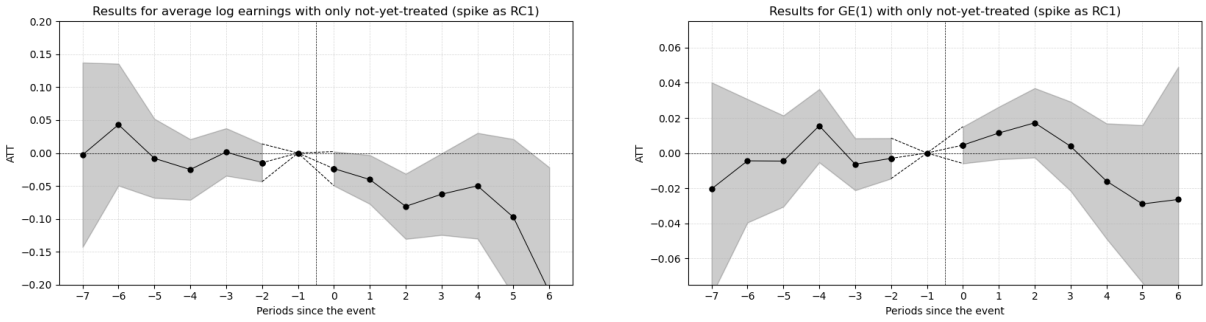

(a) Average log earnings (b) GE(1)

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the firm-level average of the logarithm of earnings in panel (a) and GE(1) in panel (b). The control group includes only not-yet-treated firms and firms with 10 or fewer workers are excluded. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

Some divergence appears also when using alternative approaches for identifying spikes. As shown in Figure 7a for average earnings, we continue to observe no evidence of differential pre-treatment trends, supporting the identification strategy. However, the post-treatment decline in average earnings becomes more pronounced. Whereas the baseline estimates indicated modest, statistically insignificant reductions, the alternative specifications now point to a more evident drop in earnings, with a statistically significant effect[11] emerging at $t + 1$. The point estimates in results for inequality, in their turn, follow the same pattern of a temporary rise seen in the baseline, although none of the post-treatment estimates remain significant, as shown in Figure 7b.

Figure 7: Automation, Earnings, and Within-Firm Inequality - Early vs Late Adopters, Alternative Spike Measure




(a) Average log earnings (b) GE(1)

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the firm-level average of the logarithm of earnings in panel (a) and GE(1) in panel (b). The control group includes only not-yet-treated firms. Spikes are identified according to the first alternative approach. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

[10] All significant results for inequality remain so with bootstrapped standard errors. For average log earnings, only the estimate for $t + 2$ remains significant.

[11] This estimate does not remain significant when using bootstrapped standard errors.

Otherwise, results of robustness checks are largely compatible with the baseline. Stratifying firms by treatment intensity does not lead to important differences for average earnings and inequality effects, as shown in Table C11 and Table C12 in the Appendix. Results for the placebo test for average earnings and inequality, shown respectively in Figure B4c and Figure B4d in the Appendix, are similar to those for employment. All estimates are again very close to zero, except for GE(1) in periods 5 and 6 after the automation event, which is again compatible with the overall trend of inequality measures through time as shown in Table 7, consisting of a continued reduction through time, with a peak in the final observation.

As discussed in Section 2.2, generalised entropy indices have a parameter $\theta$ which makes the measure more sensitive to changes at the top (bottom) of the distribution when it has a larger (smaller) value. We extend results for inequality by testing additionally GE(0) and GE(2) as outcome variables, thus capturing within-firm inequality while being more sensitive to changes at the bottom and top of the within-firm distribution, respectively. Results are very similar to those for GE(1). When only not-yet-treated firms compose the control group (cf. Figure B9 in the Appendix), both GE(0) and GE(2) show a temporary rise in point estimates, although estimates are not significant. The pattern is more pronounced in GE(2), which could be influenced by the greater volatility of that measure. As seen in the baseline for GE(1), the pattern for GE(0) and GE(2) also becomes more significant when never-treated firms are included in the control group (cf. Figure B10 in the Appendix).

In sum, results suggest that automation causes a very clear relative reduction in employment in automating firms when compared to those that haven't automated yet. This relative reduction is compatible with either job losses or hiring slowdowns, and descriptive evidence suggests the latter. When compared to the average manufacturing firm, i.e., including non-importing ones, automating firms are shown to be on a relative growth trajectory prior to automation, and that growth is shown to disappear after the automation event. While no causality can be claimed on the latter result, it suggests firms self-select into treatment by deciding to automate following a growth surge in employment. That is, the null pre-trends when automating firms are compared to firms that will automate in the future seems to be due to both sets of firms growing at a similar rate prior to automation. And the relative decline caused by automation seems to be due to the slowdown in hiring in the automating firms. These results are robust to different empirical setups and are more clearly observed for firms with automation intensity above the median.

For average earnings and within-firm inequality, results are more muted. The point estimates in baseline results suggest a small drop in average earnings and a temporary rise in inequality. Baseline coefficients are not significant, but the pattern appears more clearly in robustness checks. For instance, when excluding small firms, which may distort within-firm inequality measures, the transient rise in inequality becomes significant, as well as the reduction in average earnings. The latter also becomes clearer when an alternative approach to identifying spikes is adopted.

Taken together, results indicate that automation has led to a slowdown in firm growth, in such a way that mildly reduces average earnings and increases within-firm inequality. While overall inequality impacts would require additional cross-firm analyses, results in the literature show how within-firm inequality is the main component of inequality at the economy level, highlighting potential negative social impacts of technological change in the South African manufacturing sector in the covered period.

# 5 Conclusion

This paper investigates the effects of automation on employment dynamics and within-firm earnings inequality in South Africa's formal manufacturing sector. We apply a staggered, heterogeneity-robust

Difference-in-Differences design on rich tax administrative matched employer-employee panel data merged with firm-level import transactions.

Our findings provide evidence of selection into automation: firms adopting automation are typically on a faster employment growth trajectory prior to adoption and, in many cases, continue to expand afterwards. This appears to be the case particularly for smaller firms. However, once we account for the endogeneity of automation by comparing early and late adopters, we find that automation is associated with a relative slowdown in employment growth among automating firms following the automation episode. Importantly, this does not reflect absolute job losses, but rather a deceleration in the pace of employment expansion. While the effects on average earnings are modest and mostly statistically insignificant, our results reveal substantial heterogeneity, highlighting the distributional consequences of automation. Specifically, within-firm earnings inequality increases in the short term following automation, particularly among larger firms. This suggests that technological upgrading may reinforce within-firm earnings inequality under certain conditions. Taken together, these results highlight that while automation does not necessarily reduce employment or earnings on average, its impact on the structure of earnings inequality within firms – and potentially across the broader economy – warrants sustained empirical attention.

Further investigation is needed regarding which parts of the within-firm distribution are being impacted by the effects of automation. This is also critical to draw precise policy implications for this work. For instance, the slowdown on employment growth with increased inequality would be consistent with a hollowing-out of the middle of the within-firm earnings distribution. However, it would also be consistent with an increased hiring of high-earning workers, while low- and mid-earning ones stagnate relatively. These different scenarios would lead to diverging policy implications. A future version of this paper will include an investigation on the underlying mechanisms of the observed results that will focus on stratifying the within-firm earnings distribution to assess how workers are being impacted conditional on where they lie in that distribution.

Our approach has the important limitation of not capturing cases in which firms do not import the key capital goods themselves, but rather do so via third parties or brokers. Unfortunately, our data does not allow us to account explicitly for such cases. We mitigate this issue by restricting the main analyses to firms that have not yet automated. This means that potential firms that have not automated based on their own imports, and were thus misclassified as never-treated, are excluded from the main results. We also identify spikes using current-value imports. This is not an issue under the Bessen et al. (2023) definition, and it is a minor concern under the alternative Domini et al. (2021) definition given the low exclusion threshold, but under our main definition it may bias spike identification towards later years. The fact that results do not change substantially in robustness checks suggests that this could be only a minor issue. Beyond the scope of this paper, future research can pursue several promising directions. First, the absence of detailed information on occupations or skills in our data limits our ability to examine how automation affects workers across the task or skill distribution. Future work could leverage datasets with richer occupational classifications to assess whether rising within-firm inequality reflects increased polarization between high- and low-skilled jobs. Second, it may be valuable to investigate whether automation-induced changes at the firm level are linked to broader reallocation dynamics – for instance, whether automation enables more productive firms to grow at the expense of less productive ones. Understanding these patterns may shed light on whether automation contributes to aggregate productivity growth while deepening inequality. Finally, as more years of data become available, it will be important to examine the longer-term consequences of automation, including wage recovery or labour market transitions among displaced workers.

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# A Data cleaning

This section of the Appendix shows further details on the use of data, beyond those already explained in section 2. We use job-level data from the IRP5 datasets (National Treasury and UNU-WIDER, 2023d), individual-level identifiers from the individual panel (National Treasury and UNU-WIDER, 2023c), and firm-level data from the firm-level CIT-IRP5 panel version 4.0 (National Treasury and UNU-WIDER, 2023a) and from the customs transaction-level dataset (National Treasury and UNU-WIDER, 2023b). All datasets were accessed at the NT-SDF first in Pretoria and later in Centurion on initial visits in September 2024, and multiple visits between December 2024 and July 2025. Access was provided under a non-disclosure agreement, and the output was checked so that no firm or individual would be compromised. Results do not represent any official statistics (NT or SARS), and views expressed in the research are not necessarily the views of the National Treasury or SARS. On-site data work was done on Stata, with some post-processing on Python. Codes may be made available upon request.

Firms are identified as CIT entities (`taxrefno` or `tax_reference_number_cit`) and their sector (SIC7 Section) is identified by `imp_mic_sic7_1d`, following Budlender and Ebrahim (2020). Job-level data was loaded from the yearly IRP5 datasets, including the associated firm (CIT entity, `taxrefno`) and columns for income codes associated with wages and remuneration. We adopted all codes listed in the appendix of Kerr (2020), including 3615 "Director's remuneration" and 3616 "Independent contractor's income". The former was included because it is relevant to understand earnings at top quantiles. The latter was included following inspection of the data: some observations were identified in which individuals seemed to drop out of the data from one year to the other, but this was driven by their incomes being turned into 3616 incomes. The full list of codes is shown in Table A1. Data on all sources of income were summed for each job (`irp5it3aid`) and entries with missing `taxrefno` were dropped.

The individual identifier (`id_d_person`) associated to each job entry (`irp5it3aid`) was included by merging the employment panel within the individual panel via `irp5it3aid`. Data was then aggregated from the job level (`irp5it3aid`) to the individual level (`id_d_person`), since the same individual (`id_d_person`) may appear in more than one job (`irp5it3aid`). For cases in which such jobs are all in the same firm (`taxrefno`), all earnings were summed. For cases in which jobs are in different firms, only the main source of earnings was kept. Earnings coming from ITR12 forms were also dropped, as they could not be associated with a specific firm. As a result, we are left with a matched employer-employee dataset.

Explorations of the data, particularly of the aggregated inequality measures, showed unusual patterns likely driven by outlying observations. Three further steps were taken to deal with outliers. First, all earnings within a firm $k$ in year $t$ were added up and compared with the value of the firm's sales (`g_sales`) and labour cost (`x_labcost`) in that year. If this ratio was above 100 or below 0.01, that firm-year observation was dropped. Additionally, many workers would show deviations from their overall trend in a given year in ways that did not seem reasonable. In some cases, their earnings would drop to 1 or a very low value in a single year and then back to levels compatible with the trend. To deal with these, also following Kerr (2020), all individual-year observations with earnings below the mandatory reporting threshold of ZAR2,000 were removed. In other cases, earnings of a given individual would spike in a given year only to come back to the usual trend in the remaining years. We then deflated all earnings and calculated, for each observation of worker $i$ in year $t$, the ratio of earnings in that year over the median earning of that worker over time. Whenever this ratio was higher than 5, that individual-year observation was dropped.

Earnings and all relevant firm-level variables were deflated with the economy-wide GDP deflator (`defl_gdp_economywide`) available in the dataset, which has 2012 Q1 as the baseline. The deflator is made available as a quarterly price index, which we aggregate via the mean to a yearly measure.

As mentioned above, some firm-level variables were brought in from the firm-level panel, including the 1-digit SIC7 industry code (`imp_mic_sic7_1d`), sales (`g_sales`), cost of sales (`g_cos`), and the dormant firm dummy (`ITR14_c_CompDormant`). Inequality measures are equal to zero in firms with a single worker, so firm-year observations with a single worker were dropped. Similarly, firm-year observations with zero sales were dropped. Explorations of the firm-level data showed unusual patterns before 2014 and in 2022, leading to the choice of restricting the analysis to 2014–2021.

Other firm-level variables were aggregated from the individual-level data. These include the log number of workers, total and average log earnings paid, and the within-inequality measures discussed in section 2.2. Naturally, we calculate within-firm inequality from the list of workers assigned to each firm in a given year after all cleaning steps above. For the number of workers in the firm, we had two options. The first is counting workers assigned to the firm in a given year in our final dataset. This makes firm size more consistent with our measures of within-firm inequality, but makes the actual firm size less precise[12]. A second option is to use one of the many counts of the number of workers made available at the dataset, such as `irp5_kerr_weight_b`. To keep consistency with our within-inequality measures, analyses with firm size as outcome use the number of workers matched to the firm in our dataset.

For imports data, we used all available yearly datasets for transaction-level customs datasets between 2009 and 2022. We kept only transactions with `sa_trade_dummy` equal to 1. We then identified the imports of our focal capital goods related to automation in manufacturing via their HS2012 six-digit code (`hs6`), as found in Table 1. After dropping all other imports, we aggregated the data at the firm-year level (`taxrefno` and `taxyear`) by adding up the value (`customsvalue`) for each transaction. Spikes were then identified following the three approaches discussed in section 2.3: (i) only the highest period with import of target goods by a given firm counts as a spike; (ii) all years with imports of target goods above 20% of such imports in the highest period count as peaks, and firms with a single peak are deemed to have a spike; (iii) all years in which the ratio of imports of target goods to operation cost exceeds three times the firm-specific average of this same ratio count as peaks, and firms with a single peak are deemed to have a spike. After identifying the spikes, we merged in the previously cleaned firm-level data, restricting the import data to 2014–2021, but keeping the information for firms with spikes before and after that period.

Table A1: Income source codes used in the analysis

| Code | Description |
|---|---|
| 3601 | Income – PAYE |
| 3605 | Annual Payment – PAYE |
| 3606 | Commission – PAYE |
| 3607 | Overtime – PAYE |
| 3615 | Director's remuneration |
| 3616 | Independent contractors' income |
| 3701 | Travel allowance – PAYE |
| 3702 | Reimbursed travel allowance – PAYE |
| 3703 | Reimbursed travel allowance – IT |

[12]See Pieterse et al. (2018), Kerr (2020), and Ebrahim et al. (2021) for discussions around how to calculate the number of workers in a firm using this data

Table A1: Income source codes used in the analysis

| Code | Description |
|---|---|
| 3704 | Subsistence allowance local travel – IT |
| 3707 | Share option exercised – PAYE |
| 3708 | Public office allowance – PAYE |
| 3710 | Uniform allowance |
| 3711 | Tool allowance |
| 3712 | Acting allowance |
| 3713 | Phone allowance |
| 3714 | Other allowances – PAYE |
| 3715 | Other allowances – Excl |
| 3716 | Subsistence allowance foreign travel – IT |
| 3717 | Broad-based employee share plan – PAYE |
| 3718 | Other benefits – PAYE |
| 3719 | Other benefits – Excl |
| 3751 | Foreign bursaries – foreign service income |
| 3752 | Reimbursed travel allowance – foreign service income |
| 3753 | Travel allowance – foreign service income |
| 3754 | Share option exercised – foreign service income |
| 3755 | Other allowances – foreign service income |
| 3760 | Other non-taxable allowances – foreign service income |
| 3764 | BBE share plan – foreign service income |
| 3768 | Vesting of equity instruments – foreign service income |
| 3769 | Other fringe benefits – PAYE |
| 3801 | General fringe benefits – PAYE |
| 3802 | Use of motor vehicle acquisition by employer, not lease – PAYE |
| 3803 | Use of asset – PAYE |
| 3804 | Meals, etc. – PAYE |
| 3805 | Accommodation – PAYE |
| 3808 | Employee's debt – PAYE |
| 3809 | Taxable bursaries or scholarships – PAYE |
| 3810 | Medical aid contributions – PAYE |
| 3813 | Medical services costs – PAYE |
| 3814 | Non-taxable benefit in respect of NSF pension benefits paid by the employer |
| 3815 | Non-taxable bursaries or scholarships – Excl |
| 3816 | Use of motor vehicle acquisition by employer by lease – PAYE |
| 3820 | Taxable bursaries or scholarships (FE) – PAYE |
| 3821 | Non-taxable bursaries or scholarships (FE) – Excl |
| 3852 | Use of motor vehicle acquisition by employer, not lease – foreign income |
| 3855 | Foreign accommodation |
| 3858 | Foreign employee's debt |
| 3860 | Foreign other allowances |
| 3861 | Medical aid contributions – foreign service income |
| 3863 | Medical service costs (PAYE) – foreign service income |
| 3864 | Medical service costs – foreign service income |
| 3870 | Foreign fringe benefits – PAYE |

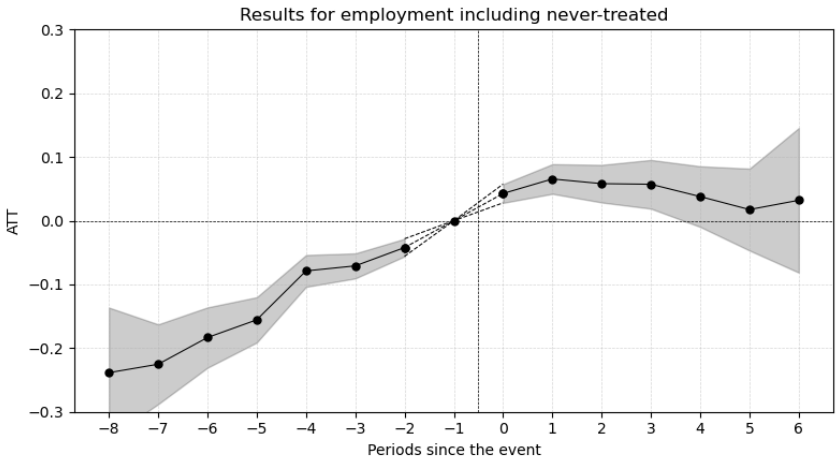
Table A1: Income source codes used in the analysis

| Code | Description |
|---|---|
| 3880 | Non-taxable bursaries or scholarships – foreign services |

Source: adapted from Kerr (2020).

# B Additional figures

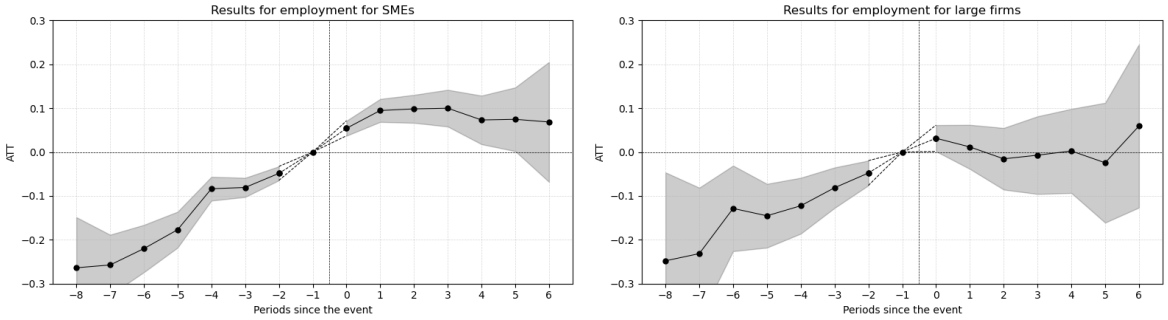
Figure B1: Automation and Firms' Total Employment - Including never-treated




Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The control group includes not-yet-treated and never-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

Figure B2: Automation and Firms' Total Employment by Firm Size Group, Including Never-Treated

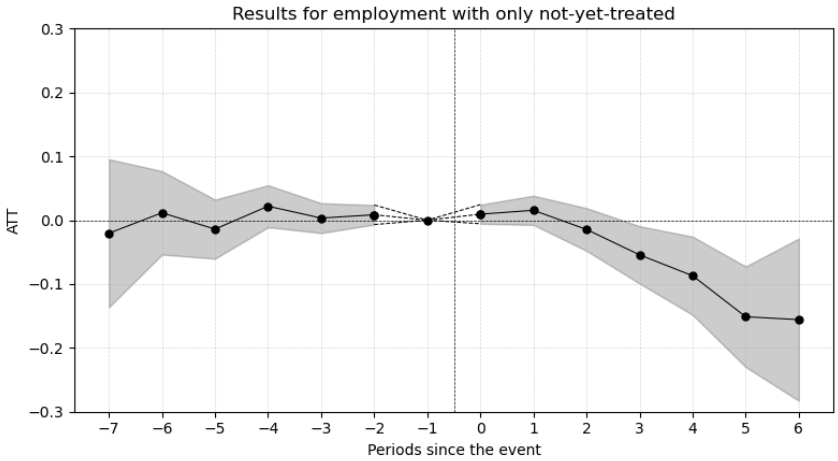



(a) Small and medium-sized firms (Size <=100) (b) Large Firms (Size >100)

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The control group includes never-treated and not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

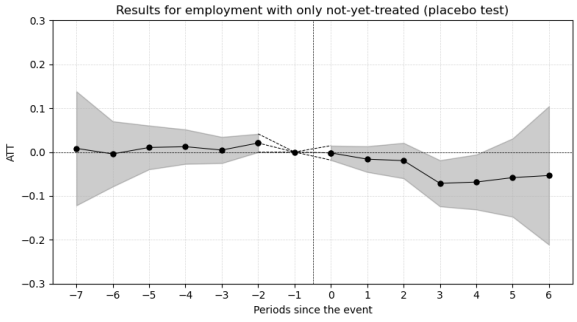
Figure B3: Automation and Firms' Total Employment - Early vs Late Adopters, Excluding Small Firms



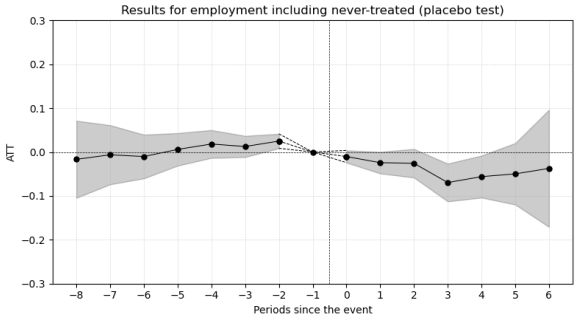

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The control group includes only not-yet-treated firms and firms with 10 or fewer workers are excluded. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

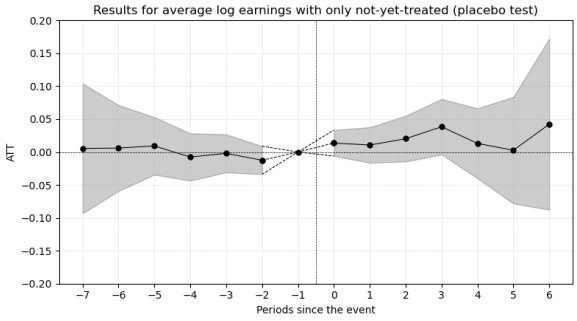
Figure B4: Automation, Employment, Earnings, and Within-Firm Inequality - Placebo Test



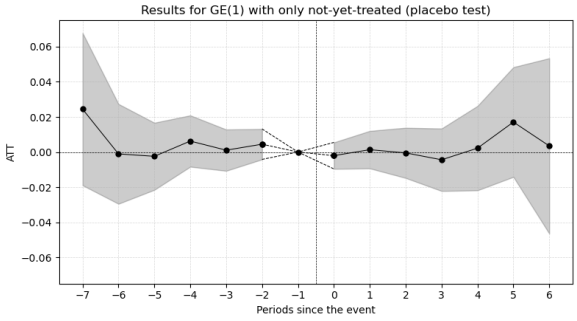

(a) Log total employment, only not-yet-treated




(b) Log total employment, including never-treated




(c) Average log earnings, only not-yet-treated

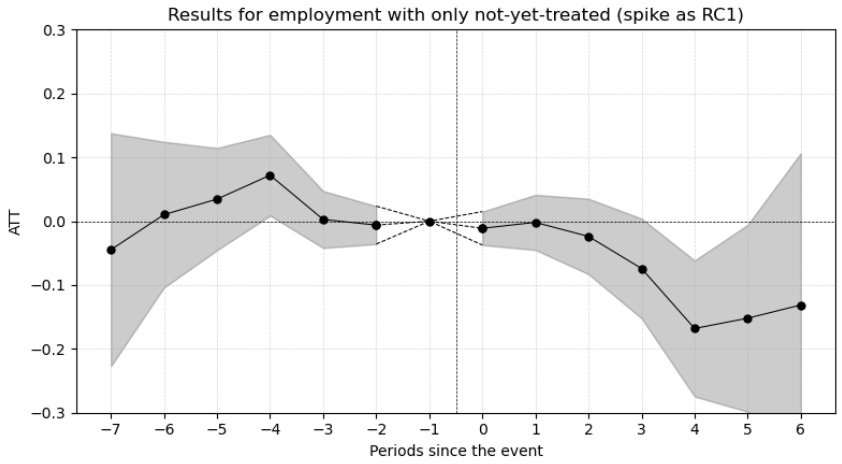



(d) GE(1), only not-yet-treated

Note: Each panel shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The control group in panels (a), (c), and (d) include only not-yet-treated firms; and in panel (b) it adds never-treated firms. The dependent variables are the logarithm of a firm's total employment (panels (a) and (b)), the firm-level average of the logarithm of earnings (panel (c)), and the GE(1) measure of within-firm earnings inequality (panel (d)). The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

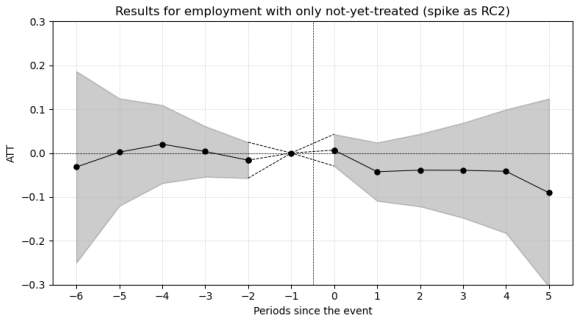
Figure B5: Automation and Firms' Total Employment - Early vs Late Adopters, Alternative Spike Measure



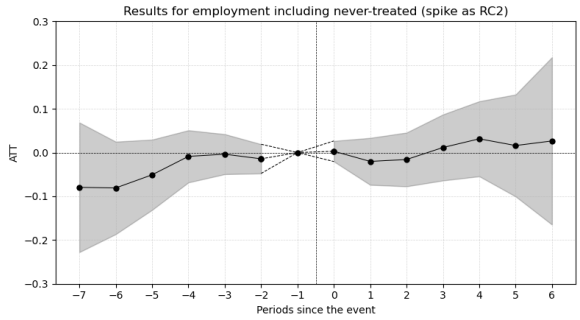

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the logarithm of a firm's total employment. The control group includes only not-yet-treated firms. Spikes are identified according to the first alternative approach. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

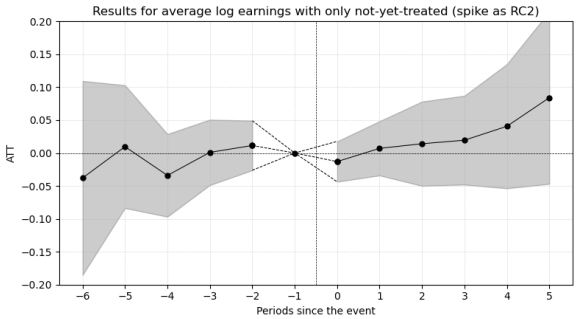
Figure B6: Automation, Employment, Earnings, and Within-Firm Inequality - Second Alternative Spike Measure



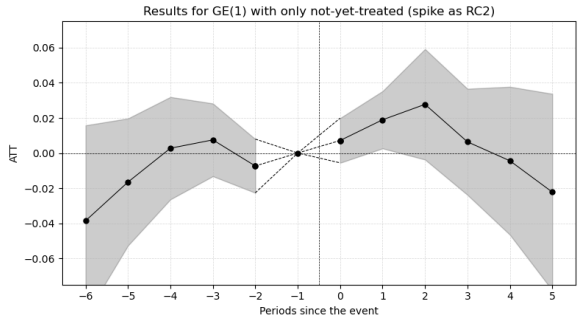

(a) Log total employment, only not-yet-treated firms




(b) Log total employment, including never-treated firms




(c) Log average earnings, only not-yet-treated firms

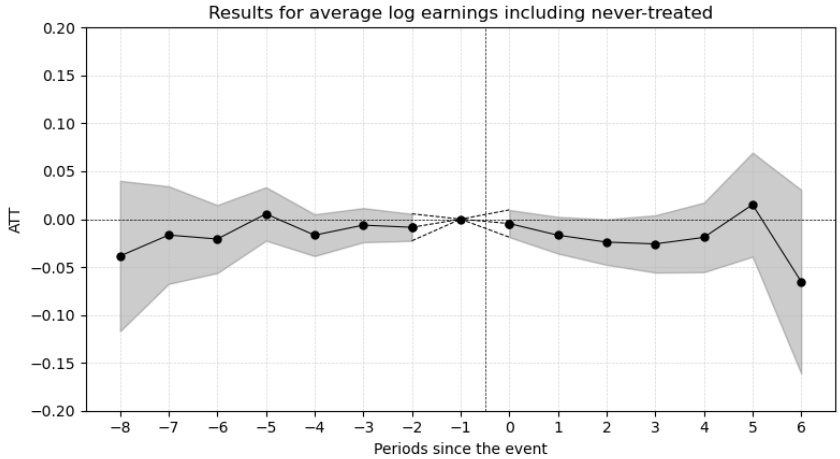



(d) Within-firm GE(1), only not-yet-treated firms

Note: Each panel shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Panels (a), (c), and (d) compare early to late adopters; panel (b) additionally includes never-treated firms in the control group. The dependent variables are the logarithm of a firm's total employment (panels (a) and (b)), the logarithm of a firm's average earnings (panel (c)), and the GE(1) measure of within-firm earnings inequality (panel (d)). Shaded areas show estimated 95% confidence intervals.

Source: authors' own calculations.

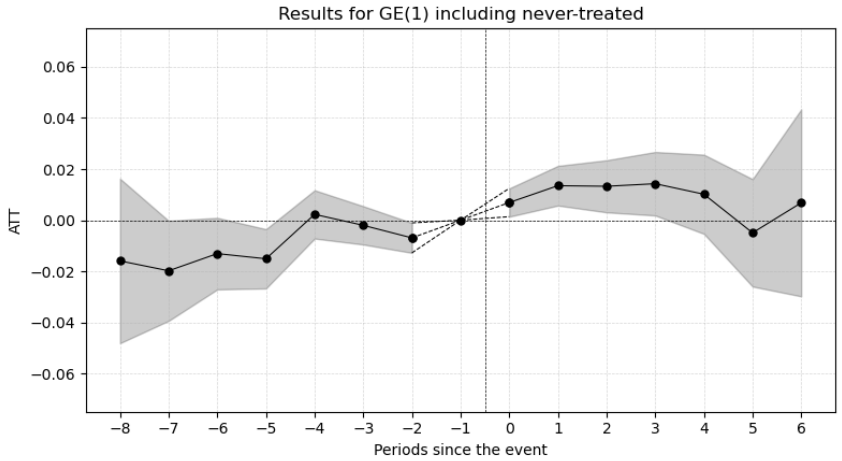
Figure B7: Automation and Firms' Average Earnings - Including Never-Treated




Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is the firm-level average of the logarithm of earnings. The control group includes never-treated and not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

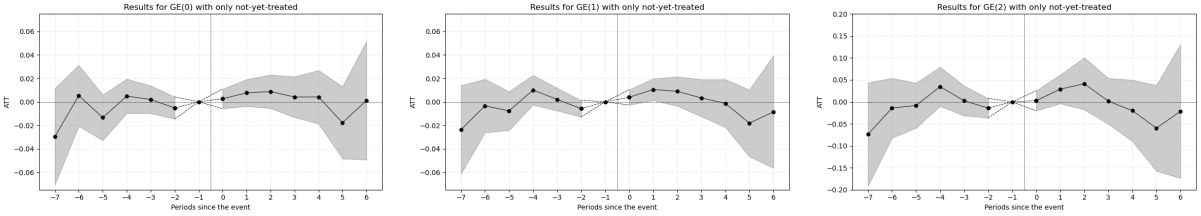
Figure B8: Automation and Within-Firm Inequality - Including Never-Treated




Note: Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is GE(1). The control group includes never-treated and not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

Figure B9: Effects of Automation on Within-Firm Inequality and Earnings - Early vs Late Adopters

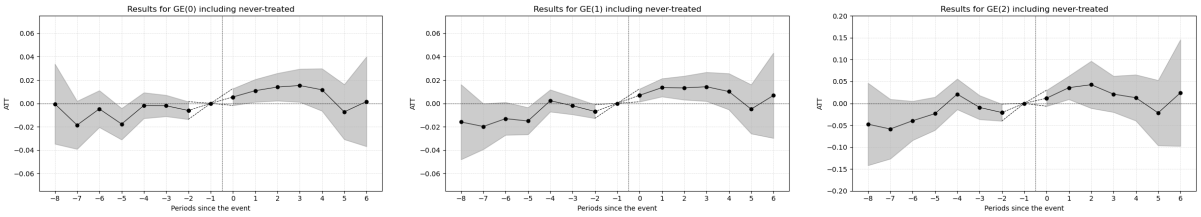



(a) GE(0) – Bottom-end inequality (b) GE(1) – Overall inequality (c) GE(2) – Top-end inequality

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is a generalized entropy measure of within-firm earnings inequality (GE(0), GE(1), or GE(2)). The control group includes only not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

Figure B10: Effects of Automation on Within-Firm Inequality and Earnings - Including Never-Treated




(a) GE(0) – Bottom-end inequality (b) GE(1) – Overall inequality (c) GE(2) – Top-end inequality

Note: The figures shows event study aggregations of group-time average treatment effects estimated with the Callaway and Sant'Anna (2021) estimator. Pre-treatment estimates use long differences. The dependent variable is a generalized entropy measure of within-firm earnings inequality (GE(0), GE(1), or GE(2)). The control group includes never-treated and not-yet-treated firms. The shaded area shows estimated 95% confidence intervals.

Source: authors' own calculations.

# C Additional tables

Table C1: Comparison between key capital goods and all imported products – number of importing firms (in all sectors) and value of imports ($10^9$ ZAR)

| Year | Importing firms | | | Value of imports | | |
|---|---|---|---|---|---|---|
| | Focal | All | Share | Focal | All | Share |
| 2009 | 8,671 | 39,656 | 21.87% | 12.69 | 560.50 | 2.26% |
| 2010 | 9,258 | 41,169 | 22.49% | 12.54 | 606.25 | 2.07% |
| 2011 | 9,410 | 41,469 | 22.69% | 14.27 | 746.28 | 1.91% |
| 2012 | 9,522 | 43,872 | 21.70% | 15.15 | 854.99 | 1.77% |
| 2013 | 9,518 | 44,290 | 21.49% | 18.42 | 998.13 | 1.85% |
| 2014 | 9,463 | 44,031 | 21.48% | 17.85 | 1,083.45 | 1.65% |
| 2015 | 9,290 | 44,843 | 20.72% | 20.26 | 1,088.00 | 1.86% |
| 2016 | 9,233 | 44,910 | 20.56% | 20.30 | 1,099.29 | 1.85% |
| 2017 | 9,871 | 47,032 | 20.99% | 19.23 | 1,106.97 | 1.74% |
| 2018 | 10,091 | 48,457 | 20.83% | 21.81 | 1,232.42 | 1.77% |
| 2019 | 9,708 | 48,089 | 20.19% | 24.43 | 1,273.31 | 1.92% |
| 2020 | 8,374 | 42,955 | 19.50% | 20.51 | 1,123.83 | 1.83% |
| 2021 | 8,592 | 40,771 | 21.07% | 22.08 | 1,380.98 | 1.60% |
| 2022 | 8,040 | 37,830 | 21.25% | 24.60 | 1,660.36 | 1.48% |

Note: values cover imports by all firms in the customs dataset, not limited to the final sample of manufacturing firms employed in the paper.

Source: authors' own calculations.

Table C2: Distribution of unique firms by firm size and treatment status for each definition of spike

| Definition of spike | Status | Small ($\leq$ 10) | Medium (11–100) | Large ($\geq$ 101) |
|---|---|---|---|---|
| Domini et al. (2022) (main) | Never-treated | 13 611 | 11 049 | 1 067 |
| | Treated | 1 863 | 3 477 | 1 390 |
| Domini et al. (2022) (alternative) | Never-treated | 13 594 | 11 051 | 1 057 |
| | Treated | 957 | 1 695 | 479 |
| | Smoothed[(a)] | 923 | 1 780 | 921 |
| Bessen et al. (2023) | Never-treated | 15 385 | *[(b)] | 2 176 |
| | Treated | 89 | 283 | 281 |
| | Smoothed | 0 | *[(b)] | 0 |
| **All firms in size group** | | **15 474** | **14 526** | **2 457** |

Note: (a) Firms with smoothed behaviour have more than one automation peak and are excluded from the analysis. (b) Omitted to comply with rules for data extraction.

Source: authors' own calculations.

Table C3: Patterns of unbalancedness in the main sample.

| Pattern | Freq. | Share | Cum. |
|---|---|---|---|
| 11111111 | 6,731 | 20.74 | 20.74 |
| 1....... | 2,055 | 6.33 | 27.07 |
| 1111111. | 1,926 | 5.93 | 33.00 |
| 11...... | 1,474 | 4.54 | 37.55 |
| ......11 | 1,120 | 3.45 | 41.00 |
| 111..... | 1,119 | 3.45 | 44.44 |
| .....111 | 1,075 | 3.31 | 47.76 |
| 1111.... | 1,010 | 3.11 | 50.87 |
| 111111.. | 976 | 3.01 | 53.87 |
| 11111... | 894 | 2.75 | 56.63 |
| .......1 | 791 | 2.44 | 59.07 |
| ....1111 | 786 | 2.42 | 61.49 |
| ..111111 | 737 | 2.27 | 63.76 |
| .1111111 | 736 | 2.27 | 66.03 |
| ......1. | 727 | 2.24 | 68.27 |
| ...11111 | 667 | 2.06 | 70.32 |
| .....1.. | 647 | 1.99 | 72.31 |
| .1...... | 536 | 1.65 | 73.97 |
| .....11. | 518 | 1.60 | 75.56 |
| ....1... | 475 | 1.46 | 77.02 |
| ..1..... | 410 | 1.26 | 78.29 |
| ...1.... | 395 | 1.22 | 79.51 |
| ....11.. | 293 | 0.90 | 80.41 |
| .11..... | 281 | 0.87 | 81.27 |
| ....111. | 273 | 0.84 | 82.11 |
| (other) | 5,805 | 17.89 | 100.00 |
| XXXXXXXX | 32,457 | 100.00 | |

Note: Patterns are described by sequences of eight symbols representing each year between 2014 and 2021. The number 1 represents an year in which the firm is present in the data, and a dot (.) represents an year in which the firm is missing.
Source: authors' own calculations.

Table C4: Yearly imports ($10^9$ ZAR) per HS4 code

| HS4 code | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | Total | Share |
|---|---|---|---|---|---|---|---|---|---|---|
| 8471 | 3.92 | 5.22 | 5.53 | 5.35 | 7.15 | 9.64 | 6.68 | 6.16 | 49.64 | 29.82% |
| 8479 | 3.20 | 3.44 | 3.02 | 2.70 | 3.49 | 2.96 | 3.15 | 3.02 | 24.98 | 15.01% |
| 8473 | 2.97 | 2.99 | 2.82 | 2.33 | 2.50 | 2.38 | 2.29 | 2.46 | 20.74 | 12.46% |
| 9032 | 1.55 | 1.69 | 1.72 | 1.50 | 1.67 | 1.58 | 1.54 | 1.73 | 12.98 | 7.80% |
| 8428 | 0.62 | 0.55 | 0.72 | 1.03 | 1.03 | 1.62 | 1.74 | 1.01 | 8.32 | 4.99% |
| 8515 | 0.65 | 0.69 | 0.97 | 0.93 | 0.66 | 1.36 | 0.82 | 1.92 | 7.99 | 4.80% |
| 8466 | 0.84 | 1.10 | 0.93 | 0.99 | 1.00 | 0.89 | 0.77 | 1.00 | 7.52 | 4.52% |
| 8462 | 0.89 | 0.89 | 0.86 | 0.76 | 0.75 | 0.75 | 0.61 | 1.35 | 6.86 | 4.12% |
| 8465 | 0.47 | 0.58 | 0.81 | 0.50 | 0.67 | 0.40 | 0.43 | 0.45 | 4.32 | 2.59% |
| 8477 | 0.56 | 0.57 | 0.50 | 0.68 | 0.50 | 0.52 | 0.55 | 0.37 | 4.25 | 2.55% |
| 8456 | 0.29 | 0.41 | 0.35 | 0.53 | 0.48 | 0.48 | 0.48 | 0.55 | 3.56 | 2.14% |
| 8458 | 0.29 | 0.38 | 0.30 | 0.25 | 0.28 | 0.26 | 0.20 | 0.22 | 2.19 | 1.31% |
| 8459 | 0.30 | 0.27 | 0.20 | 0.23 | 0.17 | 0.17 | 0.19 | 0.27 | 1.79 | 1.08% |
| 8457 | 0.18 | 0.17 | 0.31 | 0.25 | 0.14 | 0.16 | 0.10 | 0.20 | 1.50 | 0.90% |
| 8447 | 0.16 | 0.31 | 0.13 | 0.18 | 0.23 | 0.18 | 0.13 | 0.17 | 1.48 | 0.89% |
| 8470 | 0.17 | 0.17 | 0.16 | 0.17 | 0.20 | 0.20 | 0.16 | 0.16 | 1.40 | 0.84% |
| 8464 | 0.13 | 0.14 | 0.13 | 0.13 | 0.14 | 0.13 | 0.08 | 0.33 | 1.21 | 0.72% |
| 8463 | 0.18 | 0.18 | 0.14 | 0.15 | 0.11 | 0.11 | 0.08 | 0.20 | 1.15 | 0.69% |
| 8445 | 0.10 | 0.10 | 0.24 | 0.10 | 0.21 | 0.10 | 0.07 | 0.08 | 1.00 | 0.60% |
| 8461 | 0.12 | 0.14 | 0.16 | 0.14 | 0.13 | 0.11 | 0.07 | 0.12 | 0.99 | 0.59% |
| 8468 | 0.13 | 0.11 | 0.10 | 0.09 | 0.09 | 0.10 | 0.20 | 0.15 | 0.96 | 0.58% |
| 8460 | 0.09 | 0.09 | 0.07 | 0.14 | 0.09 | 0.16 | 0.08 | 0.09 | 0.81 | 0.49% |
| 8446 | 0.04 | 0.06 | 0.06 | 0.09 | 0.09 | 0.14 | 0.03 | 0.05 | 0.57 | 0.34% |
| 8444 | 0.01 | 0.01 | 0.07 | 0.02 | 0.04 | 0.02 | 0.06 | 0.02 | 0.26 | 0.16% |

Source: authors' own calculations.

Table C5: Descriptions of HS4 codes in imports

| HS4 | Description |
|---|---|
| 8471 | Automatic data processing machines and units thereof; magnetic or optical readers, machines for transcribing data onto data media in coded form and machines for processing such data, not elsewhere specified or included. |
| 8473 | Parts and accessories (other than covers, carrying cases and the like) suitable for use solely or principally with machines of headings 84.69 to 84.72. |
| 8479 | Machines and mechanical appliances having individual functions, not specified or included elsewhere in this Chapter. |
| 9032 | Automatic regulating or controlling instruments and apparatus. |
| 8515 | Electric (including electrically heated gas), laser or other light or photon beam, ultrasonic, electron beam, magnetic pulse or plasma arc soldering, brazing or welding machines and apparatus, whether or not capable of cutting; electric machines and apparatus. |
| 8462 | Machine-tools (including presses) for working metal by forging, hammering or die-stamping; machine-tools (including presses) for working metal by bending, folding, straightening, flattening, shearing, punching or notching; presses for working metal or metal parts. |
| 8466 | Parts and accessories suitable for use solely or principally with the machines of headings 84.56 to 84.65, including work or tool holders, self-opening dieheads, dividing heads and other special attachments for machine-tools; tool holders for any type of tool. |
| 8428 | Other lifting, handling, loading or unloading machinery (for example, lifts, escalators, conveyors, teleferics). |
| 8465 | Machine-tools (including machines for nailing, stapling, glueing or otherwise assembling) for working wood, cork, bone, hard rubber, hard plastics or similar hard materials. |
| 8477 | Machinery for working rubber or plastics or for the manufacture of products from these materials, not specified or included elsewhere in this Chapter. |
| 8456 | Machine-tools for working any material by removal of material, by laser or other light or photon beam, ultrasonic, electro-discharge, electro-chemical, electron beam, ionic-beam or plasma arc processes; water-jet cutting machines. |
| 8458 | Lathes (including turning centres) for removing metal. |
| 8459 | Machine-tools (including way-type unit head machines) for drilling, boring, milling, threading or tapping by removing metal, other than lathes (including turning centres) of heading 84.58. |
| 8457 | Machining centres, unit construction machines (single station) and multi-station transfer machines, for working metal. |
| 8447 | Knitting machines, stitch-bonding machines and machines for making gimped yarn, tulle, lace, embroidery, trimmings, braid or net and machines for tufting. |
| 8470 | Calculating machines and pocket-size data recording, reproducing and displaying machines with calculating functions; accounting machines, postage-franking machines, ticket-issuing machines and similar machines, incorporating a calculating device; cash registers. |
| 8463 | Other machine-tools for working metal or cermets, without removing material. |
| 8461 | Machine-tools for planing, shaping, slotting, broaching, gear cutting, gear grinding or gear finishing, sawing, cutting-off and other machine-tools working by removing metal or cermets, not elsewhere specified or included. |
| 8464 | Machine-tools for working stone, ceramics, concrete, asbestos-cement or like mineral materials or for cold working glass. |
| 8468 | Machinery and apparatus for soldering, brazing or welding, whether or not capable of cutting, other than those of heading 85.15; gas-operated surface tempering machines and appliances. |
| 8460 | Machine-tools for deburring, sharpening, grinding, honing, lapping, polishing or otherwise finishing metal or cermets by means of grinding stones, abrasives or polishing products, other than gear cutting, gear grinding or gear finishing machines of heading 84.61. |
| 8445 | Machines for preparing textile fibres; spinning, doubling or twisting machines and other machinery for producing textile yarns; textile reeling or winding (including weft-winding) machines and machines for preparing textile yarns for use on the machines of this Chapter. |
| 8446 | Weaving machines (looms). |
| 8444 | Machines for extruding, drawing, texturing or cutting man-made textile materials. |

Table C6: Event studies for the baseline (only not-yet-treated firms in the control group)

| Spike approach | | | Main | | |
|---|---|---|---|---|---|
| Sample | | | Baseline | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | -0.006 | 0.047 | -0.024 | -0.030 | -0.074 |
| | (0.062) | (0.045) | (0.019) | (0.021) | (0.060) |
| Tm6 | 0.021 | 0.008 | -0.003 | 0.005 | -0.014 |
| | (0.038) | (0.029) | (0.012) | (0.013) | (0.035) |
| Tm5 | 0.001 | 0.015 | -0.008 | -0.013 | -0.008 |
| | (0.027) | (0.020) | (0.008) | (0.010) | (0.026) |
| Tm4 | 0.042* | -0.011 | 0.010 | 0.005 | 0.035 |
| | (0.019) | (0.015) | (0.006) | (0.007) | (0.023) |
| Tm3 | 0.002 | -0.002 | 0.002 | 0.002 | 0.002 |
| | (0.013) | (0.012) | (0.005) | (0.006) | (0.017) |
| Tm2 | -0.003 | -0.005 | -0.006 | -0.005 | -0.014 |
| | (0.009) | (0.009) | (0.004) | (0.005) | (0.011) |
| Tm1 | - | - | - | - | - |
| Tp0 | 0.004 | -0.010 | 0.004 | 0.003 | 0.003 |
| | (0.008) | (0.008) | (0.003) | (0.004) | (0.012) |
| Tp1 | -0.006 | -0.018 | 0.011* | 0.008 | 0.029 |
| | (0.014) | (0.012) | (0.005) | (0.006) | (0.017) |
| Tp2 | -0.042* | -0.032* | 0.009 | 0.009 | 0.041 |
| | (0.019) | (0.015) | (0.006) | (0.007) | (0.030) |
| Tp3 | -0.093*** | -0.029 | 0.003 | 0.004 | 0.002 |
| | (0.025) | (0.020) | (0.008) | (0.009) | (0.027) |
| Tp4 | -0.145*** | -0.037 | -0.001 | 0.004 | -0.020 |
| | (0.034) | (0.025) | (0.010) | (0.012) | (0.036) |
| Tp5 | -0.197*** | -0.009 | -0.018 | -0.018 | -0.060 |
| | (0.045) | (0.037) | (0.015) | (0.016) | (0.050) |
| Tp6 | -0.197* | -0.102 | -0.009 | 0.001 | -0.021 |
| | (0.077) | (0.062) | (0.024) | (0.026) | (0.078) |
| NT in Control | NO | NO | NO | NO | NO |
| Time FE | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The baseline specification includes only not-yet-treated firms in the control group.
Source:: authors' own calculations.

Table C7: Event studies including never-treated firms in the control group

| Spike approach | Main | | | | | | |
|---|---|---|---|---|---|---|---|
| Sample | NT | NT (SME) | NT (Large) | NT | | | |
| Outcome | LNW | LNW | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm8 | -0.239*** | -0.264*** | -0.248* | -0.039 | -0.016 | -0.000 | -0.048 |
| | (0.052) | (0.059) | (0.103) | (0.040) | (0.016) | (0.018) | (0.048) |
| Tm7 | -0.225*** | -0.257*** | -0.232** | -0.017 | -0.020* | -0.019 | -0.059 |
| | (0.032) | (0.035) | (0.077) | (0.026) | (0.010) | (0.010) | (0.035) |
| Tm6 | -0.183*** | -0.220*** | -0.129* | -0.021 | -0.013 | -0.005 | -0.040 |
| | (0.024) | (0.028) | (0.050) | (0.018) | (0.007) | (0.008) | (0.023) |
| Tm5 | -0.156*** | -0.177*** | -0.145*** | 0.005 | -0.015* | -0.018* | -0.023 |
| | (0.018) | (0.021) | (0.037) | (0.014) | (0.006) | (0.007) | (0.019) |
| Tm4 | -0.079*** | -0.084*** | -0.122*** | -0.017 | 0.002 | -0.002 | 0.021 |
| | (0.013) | (0.014) | (0.033) | (0.011) | (0.005) | (0.006) | (0.018) |
| Tm3 | -0.071*** | -0.081*** | -0.081*** | -0.006 | -0.002 | -0.002 | -0.009 |
| | (0.010) | (0.011) | (0.024) | (0.009) | (0.004) | (0.005) | (0.014) |
| Tm2 | -0.042*** | -0.049*** | -0.048*** | -0.009 | -0.007* | -0.006 | -0.021* |
| | (0.007) | (0.008) | (0.014) | (0.007) | (0.003) | (0.004) | (0.010) |
| Tm1 | - | - | - | - | - | - | - |
| Tp0 | 0.043*** | 0.054*** | 0.031* | -0.005 | 0.007* | 0.005 | 0.012 |
| | (0.008) | (0.009) | (0.015) | (0.007) | (0.003) | (0.004) | (0.009) |
| Tp1 | 0.065*** | 0.095*** | 0.012 | -0.017 | 0.013*** | 0.011* | 0.036** |
| | (0.012) | (0.013) | (0.026) | (0.010) | (0.004) | (0.005) | (0.014) |
| Tp2 | 0.058*** | 0.098*** | -0.015 | -0.024* | 0.013* | 0.014* | 0.043 |
| | (0.015) | (0.016) | (0.036) | (0.012) | (0.005) | (0.006) | (0.028) |
| Tp3 | 0.057** | 0.100*** | -0.007 | -0.026 | 0.014* | 0.015* | 0.021 |
| | (0.020) | (0.021) | (0.045) | (0.015) | (0.006) | (0.007) | (0.021) |
| Tp4 | 0.038 | 0.073** | 0.002 | -0.019 | 0.010 | 0.012 | 0.013 |
| | (0.024) | (0.028) | (0.049) | (0.019) | (0.008) | (0.009) | (0.027) |
| Tp5 | 0.017 | 0.075* | -0.024 | 0.015 | -0.005 | -0.007 | -0.022 |
| | (0.033) | (0.037) | (0.070) | (0.028) | (0.011) | (0.012) | (0.038) |
| Tp6 | 0.032 | 0.068 | 0.059 | -0.065 | 0.007 | 0.002 | 0.024 |
| | (0.058) | (0.070) | (0.095) | (0.049) | (0.019) | (0.020) | (0.062) |
| NT in Control | YES | YES | YES | YES | YES | YES | YES |
| Time FE | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. NT refers to never-treated firms.

Source:: authors' own calculations.

Table C8: Event studies for robustness check excluding small firms (10 or fewer workers)

| Spike approach | Main | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample | Baseline | | | | | NT | | | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | -0.020 | 0.043 | -0.011 | -0.011 | -0.066 | -0.179*** | 0.027 | -0.022* | -0.020 | -0.070 |
| | (0.059) | (0.046) | (0.020) | (0.022) | (0.071) | (0.031) | (0.025) | (0.011) | (0.011) | (0.043) |
| Tm6 | 0.012 | -0.002 | 0.003 | 0.008 | -0.001 | -0.126*** | 0.001 | -0.013 | -0.008 | -0.042 |
| | (0.033) | (0.027) | (0.012) | (0.014) | (0.041) | (0.022) | (0.017) | (0.008) | (0.008) | (0.028) |
| Tm5 | -0.014 | 0.031 | -0.005 | -0.011 | 0.005 | -0.111*** | 0.025 | -0.015* | -0.019** | -0.017 |
| | (0.024) | (0.020) | (0.009) | (0.010) | (0.032) | (0.017) | (0.014) | (0.007) | (0.007) | (0.024) |
| Tm4 | 0.022 | -0.011 | 0.008 | 0.003 | 0.041 | -0.058*** | -0.009 | -0.001 | -0.007 | 0.022 |
| | (0.017) | (0.015) | (0.007) | (0.008) | (0.027) | (0.012) | (0.011) | (0.005) | (0.006) | (0.022) |
| Tm3 | 0.003 | 0.013 | 0.000 | -0.002 | 0.004 | -0.048*** | 0.007 | -0.004 | -0.006 | -0.006 |
| | (0.012) | (0.011) | (0.005) | (0.006) | (0.021) | (0.009) | (0.009) | (0.004) | (0.004) | (0.016) |
| Tm2 | 0.008 | -0.001 | -0.004 | -0.004 | -0.011 | -0.024*** | -0.002 | -0.005 | -0.006 | -0.018 |
| | (0.008) | (0.009) | (0.004) | (0.005) | (0.014) | (0.006) | (0.007) | (0.003) | (0.004) | (0.012) |
| Tm1 | - | - | - | - | - | - | - | - | - | - |
| Tp0 | 0.009 | -0.011 | 0.008* | 0.007 | 0.009 | 0.037*** | -0.009 | 0.010*** | 0.009** | 0.016 |
| | (0.008) | (0.008) | (0.004) | (0.004) | (0.014) | (0.007) | (0.007) | (0.003) | (0.004) | (0.012) |
| Tp1 | 0.015 | -0.019 | 0.016** | 0.012* | 0.048* | 0.062*** | -0.021* | 0.019*** | 0.017*** | 0.052** |
| | (0.012) | (0.011) | (0.005) | (0.006) | (0.020) | (0.010) | (0.009) | (0.004) | (0.005) | (0.017) |
| Tp2 | -0.015 | -0.045** | 0.015* | 0.010 | 0.066 | 0.053*** | -0.044*** | 0.021*** | 0.018** | 0.069* |
| | (0.017) | (0.014) | (0.007) | (0.007) | (0.038) | (0.014) | (0.011) | (0.006) | (0.006) | (0.035) |
| Tp3 | -0.054* | -0.035 | 0.009 | 0.007 | 0.016 | 0.045* | -0.044** | 0.021** | 0.020** | 0.036 |
| | (0.023) | (0.019) | (0.009) | (0.009) | (0.032) | (0.018) | (0.015) | (0.007) | (0.007) | (0.026) |
| Tp4 | -0.087** | -0.034 | 0.003 | 0.002 | -0.011 | 0.036 | -0.040* | 0.016 | 0.013 | 0.027 |
| | (0.031) | (0.024) | (0.011) | (0.012) | (0.043) | (0.024) | (0.019) | (0.009) | (0.010) | (0.034) |
| Tp5 | -0.151*** | -0.002 | -0.015 | -0.021 | -0.049 | 0.012 | -0.020 | 0.004 | -0.005 | 0.001 |
| | (0.040) | (0.035) | (0.015) | (0.016) | (0.060) | (0.030) | (0.026) | (0.012) | (0.012) | (0.048) |
| Tp6 | -0.156* | -0.080 | 0.007 | 0.023 | -0.005 | 0.007 | -0.090 | 0.027 | 0.029 | 0.054 |
| | (0.065) | (0.058) | (0.026) | (0.026) | (0.095) | (0.051) | (0.046) | (0.021) | (0.020) | (0.079) |
| NT in Control | NO | NO | NO | NO | NO | YES | YES | YES | YES | YES |
| Time FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The baseline specification includes only not-yet-treated firms in the control group. NT refers to never-treated firms. Firms with 10 or fewer workers are excluded.

Source:: authors' own calculations.

Table C9: Event studies for robustness check with the first alternative spike measure (RC1)

| Spike approach | Robustness check (RC1) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample | Baseline | | | | | NT | | | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | -0.045 | -0.003 | -0.020 | -0.037 | -0.031 | -0.232*** | -0.049 | -0.036* | -0.032* | -0.074 |
| | (0.093) | (0.072) | (0.031) | (0.034) | (0.083) | (0.048) | (0.039) | (0.015) | (0.016) | (0.044) |
| Tm6 | 0.010 | 0.043 | -0.004 | 0.002 | 0.000 | -0.210*** | -0.020 | -0.024* | -0.014 | -0.039 |
| | (0.058) | (0.047) | (0.018) | (0.020) | (0.055) | (0.038) | (0.029) | (0.011) | (0.012) | (0.033) |
| Tm5 | 0.035 | -0.008 | -0.005 | -0.010 | 0.010 | -0.151*** | -0.030 | -0.012 | -0.013 | 0.002 |
| | (0.041) | (0.031) | (0.013) | (0.015) | (0.040) | (0.027) | (0.022) | (0.009) | (0.010) | (0.027) |
| Tm4 | 0.072* | -0.025 | 0.016 | 0.012 | 0.041 | -0.068*** | -0.046** | 0.010 | 0.007 | 0.042 |
| | (0.032) | (0.024) | (0.011) | (0.012) | (0.039) | (0.020) | (0.017) | (0.008) | (0.009) | (0.029) |
| Tm3 | 0.003 | 0.002 | -0.006 | -0.004 | -0.032 | -0.083*** | -0.015 | -0.006 | -0.003 | -0.026 |
| | (0.023) | (0.018) | (0.008) | (0.009) | (0.022) | (0.017) | (0.014) | (0.006) | (0.007) | (0.020) |
| Tm2 | -0.006 | -0.015 | -0.003 | -0.001 | -0.003 | -0.054*** | -0.026* | -0.005 | -0.001 | -0.016 |
| | (0.015) | (0.015) | (0.006) | (0.008) | (0.017) | (0.012) | (0.011) | (0.005) | (0.006) | (0.017) |
| Tm1 | - | - | - | - | - | - | - | - | - | - |
| Tp0 | -0.011 | -0.024 | 0.005 | 0.009 | 0.001 | 0.037** | -0.007 | 0.008 | 0.011 | 0.011 |
| | (0.013) | (0.013) | (0.005) | (0.007) | (0.016) | (0.012) | (0.011) | (0.004) | (0.006) | (0.011) |
| Tp1 | -0.002 | -0.040* | 0.011 | 0.009 | 0.021 | 0.079*** | -0.025 | 0.014* | 0.010 | 0.024 |
| | (0.022) | (0.019) | (0.008) | (0.009) | (0.026) | (0.018) | (0.016) | (0.006) | (0.008) | (0.019) |
| Tp2 | -0.024 | -0.081** | 0.017 | 0.017 | 0.057 | 0.084*** | -0.048* | 0.018* | 0.017 | 0.040 |
| | (0.030) | (0.025) | (0.010) | (0.012) | (0.034) | (0.024) | (0.020) | (0.008) | (0.010) | (0.027) |
| Tp3 | -0.074 | -0.063* | 0.004 | -0.001 | 0.056 | 0.098*** | -0.036 | 0.016 | 0.009 | 0.064* |
| | (0.040) | (0.032) | (0.013) | (0.015) | (0.037) | (0.029) | (0.025) | (0.010) | (0.012) | (0.028) |
| Tp4 | -0.168** | -0.050 | -0.016 | -0.019 | 0.014 | 0.039 | -0.013 | 0.005 | -0.004 | 0.050 |
| | (0.054) | (0.041) | (0.017) | (0.020) | (0.049) | (0.039) | (0.031) | (0.013) | (0.016) | (0.035) |
| Tp5 | -0.152* | -0.098 | -0.029 | -0.032 | -0.025 | 0.062 | -0.057 | 0.001 | -0.009 | 0.042 |
| | (0.075) | (0.061) | (0.023) | (0.026) | (0.064) | (0.058) | (0.049) | (0.018) | (0.022) | (0.043) |
| Tp6 | -0.131 | -0.211* | -0.026 | -0.030 | 0.001 | 0.059 | -0.167* | 0.007 | -0.014 | 0.075 |
| | (0.121) | (0.096) | (0.039) | (0.041) | (0.120) | (0.093) | (0.081) | (0.031) | (0.033) | (0.095) |
| NT in Control | NO | NO | NO | NO | NO | YES | YES | YES | YES | YES |
| Time FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The baseline specification includes only not-yet-treated firms in the control group. NT refers to never-treated firms. Spikes are identified according to the first alternative approach.

Source:: authors' own calculations.

Table C10: Event studies for robustness check with the second alternative spike measure (RC2)

| Spike approach | Robustness check (RC2) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample | Baseline | | | | | NT | | | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm6 | -0.032 | -0.038 | -0.039 | -0.016 | -0.038 | -0.081 | 0.029 | 0.001 | -0.001 | 0.035 |
| | (0.111) | (0.075) | (0.028) | (0.037) | (0.095) | (0.054) | (0.034) | (0.015) | (0.017) | (0.051) |
| Tm5 | 0.002 | 0.010 | -0.017 | -0.027 | 0.010 | -0.051 | 0.018 | 0.005 | -0.008 | 0.062 |
| | (0.063) | (0.048) | (0.018) | (0.024) | (0.068) | (0.041) | (0.027) | (0.011) | (0.014) | (0.046) |
| Tm4 | 0.020 | -0.034 | 0.003 | 0.001 | -0.013 | -0.009 | -0.016 | 0.012 | 0.001 | 0.046 |
| | (0.045) | (0.032) | (0.015) | (0.018) | (0.046) | (0.030) | (0.023) | (0.010) | (0.012) | (0.034) |
| Tm3 | 0.003 | 0.001 | 0.007 | 0.006 | 0.072 | -0.004 | 0.006 | 0.014 | 0.007 | 0.086 |
| | (0.029) | (0.025) | (0.011) | (0.013) | (0.071) | (0.023) | (0.019) | (0.007) | (0.010) | (0.049) |
| Tm2 | -0.016 | 0.011 | -0.007 | -0.008 | -0.006 | -0.014 | 0.011 | 0.000 | -0.002 | 0.024 |
| | (0.021) | (0.019) | (0.008) | (0.011) | (0.025) | (0.017) | (0.015) | (0.006) | (0.008) | (0.020) |
| Tm1 | - | - | - | - | - | - | - | - | - | - |
| Tp0 | 0.007 | -0.013 | 0.007 | 0.004 | 0.031 | 0.003 | -0.007 | 0.003 | 0.001 | 0.011 |
| | (0.019) | (0.016) | (0.007) | (0.009) | (0.022) | (0.012) | (0.012) | (0.005) | (0.007) | (0.017) |
| Tp1 | -0.043 | 0.007 | 0.019* | 0.011 | 0.082* | -0.020 | 0.009 | 0.009 | 0.004 | 0.041 |
| | (0.034) | (0.021) | (0.008) | (0.010) | (0.038) | (0.027) | (0.017) | (0.007) | (0.008) | (0.028) |
| Tp2 | -0.039 | 0.014 | 0.028 | 0.025 | 0.192 | -0.016 | 0.017 | 0.022 | 0.022* | 0.167 |
| | (0.042) | (0.033) | (0.016) | (0.014) | (0.129) | (0.031) | (0.025) | (0.013) | (0.011) | (0.111) |
| Tp3 | -0.039 | 0.019 | 0.006 | 0.008 | 0.001 | 0.012 | -0.004 | 0.001 | 0.009 | -0.031 |
| | (0.055) | (0.034) | (0.015) | (0.018) | (0.046) | (0.039) | (0.026) | (0.012) | (0.013) | (0.035) |
| Tp4 | -0.042 | 0.041 | -0.004 | 0.001 | -0.036 | 0.031 | -0.003 | 0.003 | 0.013 | -0.034 |
| | (0.072) | (0.048) | (0.022) | (0.024) | (0.061) | (0.044) | (0.032) | (0.015) | (0.016) | (0.037) |
| Tp5 | -0.091 | 0.084 | -0.022 | -0.011 | -0.112 | 0.016 | 0.072 | -0.021 | -0.009 | -0.099* |
| | (0.110) | (0.067) | (0.029) | (0.032) | (0.078) | (0.059) | (0.037) | (0.019) | (0.020) | (0.047) |
| NT in Control | NO | NO | NO | NO | NO | YES | YES | YES | YES | YES |
| Time FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The baseline specification includes only not-yet-treated firms in the control group. NT refers to never-treated firms. Spikes are identified according to the second alternative approach.

Source:: authors' own calculations.

Table C11: Event studies for firms with above-median automation spikes

| Spike approach | Main | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample | High spike (Large) | | | | | High spike (SME) | | | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | 0.005 | 0.023 | -0.040 | -0.027 | -0.172 | 0.011 | 0.106 | -0.051 | -0.057 | -0.121 |
| | (0.137) | (0.101) | (0.042) | (0.048) | (0.186) | (0.111) | (0.071) | (0.031) | (0.033) | (0.080) |
| Tm6 | 0.025 | 0.009 | -0.029 | -0.032 | -0.040 | 0.007 | 0.033 | -0.023 | -0.007 | -0.079 |
| | (0.073) | (0.055) | (0.028) | (0.032) | (0.118) | (0.070) | (0.045) | (0.020) | (0.022) | (0.054) |
| Tm5 | -0.037 | 0.056 | -0.042* | -0.051* | -0.039 | -0.012 | 0.019 | 0.003 | 0.004 | -0.020 |
| | (0.052) | (0.041) | (0.019) | (0.021) | (0.094) | (0.053) | (0.036) | (0.015) | (0.017) | (0.036) |
| Tm4 | -0.021 | -0.031 | 0.001 | 0.002 | 0.053 | 0.074* | -0.006 | 0.019 | 0.007 | 0.060 |
| | (0.039) | (0.029) | (0.015) | (0.017) | (0.086) | (0.032) | (0.024) | (0.011) | (0.012) | (0.032) |
| Tm3 | -0.010 | 0.026 | -0.010 | -0.019 | 0.002 | 0.014 | -0.012 | 0.015 | 0.019 | 0.027 |
| | (0.028) | (0.022) | (0.010) | (0.011) | (0.074) | (0.023) | (0.020) | (0.008) | (0.010) | (0.021) |
| Tm2 | -0.000 | 0.001 | -0.010 | -0.014 | -0.040 | 0.004 | -0.008 | -0.002 | 0.003 | -0.011 |
| | (0.021) | (0.018) | (0.007) | (0.009) | (0.044) | (0.017) | (0.015) | (0.006) | (0.008) | (0.015) |
| Tm1 | - | - | - | - | - | - | - | - | - | - |
| Tp0 | 0.000 | 0.006 | 0.003 | 0.007 | -0.027 | 0.009 | -0.034* | 0.009 | 0.010 | 0.012 |
| | (0.021) | (0.017) | (0.007) | (0.008) | (0.046) | (0.015) | (0.015) | (0.006) | (0.007) | (0.014) |
| Tp1 | -0.037 | 0.022 | 0.014 | 0.010 | 0.044 | 0.000 | -0.059** | 0.018* | 0.019 | 0.039* |
| | (0.033) | (0.022) | (0.010) | (0.010) | (0.069) | (0.025) | (0.020) | (0.008) | (0.010) | (0.019) |
| Tp2 | -0.123** | 0.029 | 0.010 | 0.022 | 0.084 | -0.042 | -0.054* | 0.010 | 0.005 | 0.046 |
| | (0.047) | (0.035) | (0.018) | (0.017) | (0.147) | (0.033) | (0.025) | (0.010) | (0.012) | (0.029) |
| Tp3 | -0.153* | 0.073 | -0.014 | -0.006 | -0.129 | -0.097* | -0.060 | 0.012 | 0.009 | 0.045 |
| | (0.061) | (0.047) | (0.020) | (0.020) | (0.099) | (0.044) | (0.032) | (0.014) | (0.016) | (0.038) |
| Tp4 | -0.151* | 0.087 | -0.012 | -0.001 | -0.176 | -0.146* | -0.116** | 0.009 | 0.017 | 0.032 |
| | (0.069) | (0.052) | (0.025) | (0.026) | (0.140) | (0.061) | (0.042) | (0.017) | (0.020) | (0.046) |
| Tp5 | -0.276** | 0.203* | -0.024 | -0.004 | -0.228 | -0.209** | -0.080 | 0.007 | -0.013 | 0.059 |
| | (0.098) | (0.087) | (0.038) | (0.036) | (0.194) | (0.079) | (0.061) | (0.025) | (0.027) | (0.070) |
| Tp6 | -0.313 | 0.105 | 0.059 | 0.051 | 0.207 | -0.244 | -0.142 | 0.011 | 0.013 | 0.043 |
| | (0.164) | (0.151) | (0.069) | (0.060) | (0.300) | (0.139) | (0.100) | (0.039) | (0.039) | (0.099) |
| NT in Control | NO | NO | NO | NO | NO | NO | NO | NO | NO | NO |
| Time FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The control group includes only not-yet-treated firms.

Source:: authors' own calculations.

Table C12: Event studies for firms with below-median automation spikes

| Spike approach | Main | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample | Low spike (Large) | | | | | Low spike (SME) | | | | |
| Outcome | LNW | ALE | GE(1) | GE(0) | GE(2) | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | -0.205 | -0.141 | 0.074 | 0.032 | 0.148 | -0.024 | 0.030 | -0.008 | -0.015 | -0.031 |
| | (0.263) | (0.197) | (0.091) | (0.081) | (0.403) | (0.087) | (0.075) | (0.031) | (0.035) | (0.076) |
| Tm6 | 0.059 | -0.089 | 0.079 | 0.084* | 0.113 | 0.006 | -0.000 | 0.011 | 0.022 | 0.030 |
| | (0.121) | (0.091) | (0.046) | (0.037) | (0.230) | (0.055) | (0.053) | (0.018) | (0.021) | (0.041) |
| Tm5 | -0.100 | 0.040 | 0.026 | 0.010 | 0.028 | 0.023 | -0.010 | -0.010 | -0.017 | 0.000 |
| | (0.084) | (0.081) | (0.039) | (0.044) | (0.186) | (0.037) | (0.032) | (0.013) | (0.016) | (0.032) |
| Tm4 | -0.169 | 0.012 | 0.013 | 0.005 | -0.062 | 0.060* | -0.010 | 0.003 | 0.003 | 0.008 |
| | (0.122) | (0.067) | (0.032) | (0.037) | (0.155) | (0.026) | (0.024) | (0.010) | (0.012) | (0.022) |
| Tm3 | -0.066 | -0.005 | 0.009 | -0.002 | -0.068 | -0.001 | -0.003 | -0.004 | -0.003 | -0.010 |
| | (0.067) | (0.049) | (0.022) | (0.027) | (0.093) | (0.020) | (0.019) | (0.008) | (0.010) | (0.018) |
| Tm2 | 0.018 | -0.026 | 0.011 | 0.009 | 0.009 | -0.012 | -0.003 | -0.010 | -0.011 | -0.012 |
| | (0.028) | (0.035) | (0.015) | (0.020) | (0.062) | (0.013) | (0.014) | (0.006) | (0.008) | (0.013) |
| Tm1 | - | - | - | - | - | - | - | - | - | - |
| Tp0 | -0.043 | 0.043 | 0.003 | -0.017 | 0.026 | 0.011 | -0.005 | 0.000 | -0.002 | 0.002 |
| | (0.029) | (0.027) | (0.013) | (0.018) | (0.070) | (0.012) | (0.013) | (0.005) | (0.007) | (0.014) |
| Tp1 | -0.075 | 0.077 | 0.012 | -0.026 | 0.103 | 0.019 | -0.014 | 0.004 | 0.003 | 0.005 |
| | (0.051) | (0.045) | (0.022) | (0.026) | (0.084) | (0.020) | (0.019) | (0.008) | (0.010) | (0.021) |
| Tp2 | -0.009 | 0.064 | -0.010 | -0.045 | 0.064 | 0.003 | -0.055* | 0.012 | 0.016 | 0.023 |
| | (0.062) | (0.054) | (0.021) | (0.027) | (0.094) | (0.027) | (0.025) | (0.010) | (0.012) | (0.023) |
| Tp3 | 0.017 | 0.027 | -0.004 | -0.043 | 0.145 | -0.056 | -0.057 | 0.008 | 0.014 | 0.013 |
| | (0.077) | (0.066) | (0.028) | (0.028) | (0.134) | (0.037) | (0.032) | (0.012) | (0.014) | (0.031) |
| Tp4 | -0.036 | -0.001 | -0.009 | -0.034 | 0.048 | -0.134** | -0.027 | -0.001 | 0.002 | 0.005 |
| | (0.105) | (0.088) | (0.036) | (0.037) | (0.174) | (0.050) | (0.043) | (0.017) | (0.019) | (0.043) |
| Tp5 | -0.027 | 0.068 | -0.023 | -0.056 | 0.180 | -0.137* | -0.057 | -0.029 | -0.019 | -0.099* |
| | (0.158) | (0.119) | (0.050) | (0.046) | (0.302) | (0.068) | (0.060) | (0.021) | (0.026) | (0.050) |
| Tp6 | 0.199 | -0.089 | -0.031 | -0.049 | 0.005 | -0.137 | -0.156 | -0.043 | -0.017 | -0.155 |
| | (0.144) | (0.146) | (0.075) | (0.062) | (0.600) | (0.117) | (0.102) | (0.039) | (0.047) | (0.081) |
| NT in Control | NO | NO | NO | NO | NO | NO | NO | NO | NO | NO |
| Time FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The control group includes only not-yet-treated firms.

Source:: authors' own calculations.

Table C13: Event studies for placebo test with random treatment assignment

| Spike approach | | | Placebo | | | |
|---|---|---|---|---|---|---|
| Sample | Baseline | NT | Baseline | | | |
| Outcome | LNW | LNW | ALE | GE(1) | GE(0) | GE(2) |
| Tm7 | 0.008 | -0.006 | 0.005 | 0.024 | 0.012 | 0.098 |
| | (0.066) | (0.034) | (0.050) | (0.022) | (0.023) | (0.070) |
| Tm6 | -0.004 | -0.010 | 0.006 | -0.001 | -0.008 | 0.037 |
| | (0.038) | (0.025) | (0.033) | (0.015) | (0.016) | (0.048) |
| Tm5 | 0.010 | 0.006 | 0.009 | -0.002 | -0.007 | 0.022 |
| | (0.025) | (0.019) | (0.022) | (0.010) | (0.011) | (0.030) |
| Tm4 | 0.012 | 0.018 | -0.008 | 0.006 | 0.007 | 0.018 |
| | (0.020) | (0.016) | (0.018) | (0.007) | (0.008) | (0.025) |
| Tm3 | 0.005 | 0.013 | -0.002 | 0.001 | 0.001 | 0.008 |
| | (0.015) | (0.012) | (0.015) | (0.006) | (0.007) | (0.020) |
| Tm2 | 0.021 | 0.025** | -0.012 | 0.004 | 0.002 | 0.005 |
| | (0.011) | (0.008) | (0.011) | (0.004) | (0.005) | (0.014) |
| Tm1 | - | - | - | - | - | - |
| Tp0 | -0.002 | -0.010 | 0.014 | -0.002 | -0.006 | -0.007 |
| | (0.008) | (0.007) | (0.010) | (0.004) | (0.005) | (0.011) |
| Tp1 | -0.016 | -0.024 | 0.011 | 0.001 | 0.001 | 0.004 |
| | (0.015) | (0.013) | (0.014) | (0.005) | (0.006) | (0.017) |
| Tp2 | -0.020 | -0.026 | 0.020 | -0.000 | 0.005 | -0.020 |
| | (0.021) | (0.017) | (0.018) | (0.007) | (0.008) | (0.026) |
| Tp3 | -0.071** | -0.070** | 0.038 | -0.004 | 0.000 | -0.034 |
| | (0.027) | (0.022) | (0.022) | (0.009) | (0.010) | (0.030) |
| Tp4 | -0.068* | -0.056* | 0.013 | 0.002 | 0.017 | -0.064 |
| | (0.032) | (0.024) | (0.027) | (0.012) | (0.014) | (0.039) |
| Tp5 | -0.058 | -0.050 | 0.003 | 0.017 | 0.034 | -0.079 |
| | (0.046) | (0.036) | (0.041) | (0.016) | (0.018) | (0.054) |
| Tp6 | -0.053 | -0.037 | 0.042 | 0.003 | 0.022 | -0.119 |
| | (0.081) | (0.068) | (0.066) | (0.026) | (0.029) | (0.086) |
| NT in Control | NO | YES | NO | NO | NO | NO |
| Time FE | YES | YES | YES | YES | YES | YES |
| Firm FE | YES | YES | YES | YES | YES | YES |

Note: LNW refers to the log number of workers. ALE refers to average log earnings. The baseline specification includes only not-yet-treated firms in the control group. NT refers to never-treated firms.

Source:: authors' own calculations.